Elementary Analysis

Chapter 2: Sections 9–10

All 30 exercises, with every subpart
Detailed worked solutions

Mathjock

Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Chapter 2. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, sequence indices run through $\mathbb{N}$ and all terms and finite limits are real. The empty sum is 0, and $0! = 1$.

Coverage. Section 9: Exercises 9.1–9.18. Section 10: Exercises 10.1–10.12.

Contents

9 Limit theorems for sequences **3**

9.1 Exercise 9.1: Three quotients evaluated by the finite limit laws . . . 3
9.2 Exercise 9.2: Limits obtained from two given convergent sequences . . . 5
9.3 Exercise 9.3: A polynomial expression in convergent sequences . . . 5
9.4 Exercise 9.4: A nested-square-root recursion . . . 6
9.5 Exercise 9.5: A positive rational recursion . . . 8
9.6 Exercise 9.6: A necessary finite-limit equation and an actual infinite limit . . . 9
9.7 Exercise 9.7: Finish the estimate for the root of the index . . . 10
9.8 Exercise 9.8: Finite and infinite possibilities for five limits . . . 11
9.9 Exercise 9.9: Order preserved by existing extended limits . . . 13
9.10 Exercise 9.10: Scaling and reversing an infinite limit . . . 14
9.11 Exercise 9.11: Adding a sequence that is bounded below . . . 15
9.12 Exercise 9.12: The ratio test for the size of a sequence . . . 16

9.13 Exercise 9.13: All real bases in a geometric sequence 18
9.14 Exercise 9.14: Geometric growth compared with a real power 19
9.15 Exercise 9.15: Every fixed geometric sequence is dominated by a factorial 20
9.16 Exercise 9.16: Three applications of the infinite-limit results 21
9.17 Exercise 9.17: A definition-only proof for the sequence of squares 22
9.18 Exercise 9.18: Finite geometric sums and their limits 23

10 Monotone sequences and Cauchy sequences 25

10.1 Exercise 10.1: Classifying six sequences 25
10.2 Exercise 10.2: Convergence of a bounded decreasing sequence 28
10.3 Exercise 10.3: A finite bound for decimal approximations 29
10.4 Exercise 10.4: Why the monotone and Cauchy theorems require real completeness . 30
10.5 Exercise 10.5: An unbounded decreasing sequence tends to negative infinity 32
10.6 Exercise 10.6: Small successive differences: one sufficient rate and one insufficient rate 32
10.7 Exercise 10.7: Approaching an unattained supremum by points of the set 34
10.8 Exercise 10.8: Averages of an increasing positive sequence 35
10.9 Exercise 10.9: A decreasing recurrence with a squared term 36
10.10 Exercise 10.10: A linear recurrence approaching one half 38
10.11 Exercise 10.11: An infinite-product recurrence 39
10.12 Exercise 10.12: A product recurrence with an exact formula 42

9 Limit theorems for sequences

This section proves and applies rules for limits. Each application checks the hypotheses: the sequences must have the required finite limits, and a quotient needs a nonzero limiting denominator.

Definition 9.8 extends the phrase *a limit exists* to include divergence to $+\infty$ or $-\infty$. These are not finite real limits. We use the extended convention where the exercise permits it, while checking separately whether a finite-limit theorem applies. A recurrence's fixed-point equation gives a necessary condition for a finite limit; it does not prove convergence.

9.1 Exercise 9.1: Three quotients evaluated by the finite limit laws

What we must show. Use Theorems 9.2–9.7, with their hypotheses checked, to prove

$$\text{(a)} \quad \lim_{n\to\infty} \frac{n+1}{n} = 1,$$
$$\text{(b)} \quad \lim_{n\to\infty} \frac{3n+7}{6n-5} = \frac{1}{2},$$
$$\text{(c)} \quad \lim_{n\to\infty} \frac{17n^5+73n^4-18n^2+3}{23n^5+13n^3} = \frac{17}{23}.$$

Plan. Divide each polynomial expression by the appropriate positive power of n. The resulting expressions involve constants and reciprocal powers, whose limits are known. For each use of the quotient theorem, verify that the denominator terms are nonzero and that their finite limit is nonzero.

1. Record the finite limit rules being used

A constant sequence with value $c \in \mathbb{R}$ converges to c, because $|c-c| = 0 < \varepsilon$ for every $\varepsilon > 0$. Theorem 9.7(a) gives

$$\frac{1}{n^p} \longrightarrow 0 \qquad (p > 0).$$

The scalar rule is Theorem 9.2, the sum rule is Theorem 9.3, and the product rule is Theorem 9.4. Subtraction follows by combining the scalar rule with scalar -1 and the sum rule. The quotient rule, Theorem 9.6, requires a convergent numerator, a convergent denominator, a nonzero denominator at every index, and a nonzero denominator limit. All limits used in these rules are finite real numbers.

2. Part (a): Separate the constant term

Since $n \in \mathbb{N}$, the denominator n is positive. The identity

$$\frac{n+1}{n} = \frac{n}{n} + \frac{1}{n} = 1 + \frac{1}{n}$$

therefore holds for every index. The constant sequence 1 converges to 1, and Theorem 9.7(a), with $p = 1$, gives $1/n \to 0$. Applying Theorem 9.3 yields

$$\frac{n+1}{n} = 1 + \frac{1}{n} \longrightarrow 1 + 0 = 1.$$

Here the sum rule applies after rewriting; there is no need to treat the unbounded sequence n as a finite convergent sequence.

3. Part (b): Normalize numerator and denominator

For every $n \geq 1$, the original denominator satisfies $6n - 5 \geq 1 > 0$. Dividing numerator and denominator by $n > 0$ gives

$$\frac{3n+7}{6n-5} = \frac{(3n+7)/n}{(6n-5)/n} = \frac{3+7/n}{6-5/n}.$$

Let $u_n = 3+7/n$ and $v_n = 6-5/n$. By Theorem 9.7(a), $1/n \to 0$. The scalar rule gives $7/n \to 7 \cdot 0 = 0$ and $-5/n \to -5 \cdot 0 = 0$. Adding the constant sequences by Theorem 9.3 gives

$$u_n \longrightarrow 3 + 0 = 3, \qquad v_n \longrightarrow 6 + 0 = 6.$$

For each n, we also have

$$v_n = \frac{6n-5}{n} > 0.$$

Thus every denominator term is nonzero, and its limit 6 is nonzero. All hypotheses of Theorem 9.6 have now been verified, so

$$\frac{3n+7}{6n-5} = \frac{u_n}{v_n} \longrightarrow \frac{3}{6} = \frac{1}{2}.$$

4. Part (c): Divide by the largest power in the denominator

The original denominator is positive for every index:

$$23n^5 + 13n^3 = n^3(23n^2 + 13) > 0,$$

because $n \geq 1$. Divide both numerator and denominator by $n^5 > 0$. Each term then simplifies as follows:

$$\frac{n^4}{n^5} = \frac{1}{n}, \qquad \frac{n^2}{n^5} = \frac{1}{n^3}, \qquad \frac{n^3}{n^5} = \frac{1}{n^2}.$$

Consequently

$$\frac{17n^5 + 73n^4 - 18n^2 + 3}{23n^5 + 13n^3} = \frac{17 + 73/n - 18/n^3 + 3/n^5}{23 + 13/n^2}.$$

Write the normalized numerator and denominator as U_n and V_n. Theorem 9.7(a), applied with $p = 1, 2, 3, 5$, gives

$$\frac{1}{n} \to 0, \qquad \frac{1}{n^2} \to 0, \qquad \frac{1}{n^3} \to 0, \qquad \frac{1}{n^5} \to 0.$$

Multiplication by the indicated constants, using Theorem 9.2, gives

$$\frac{73}{n} \to 0, \qquad -\frac{18}{n^3} \to 0, \qquad \frac{3}{n^5} \to 0, \qquad \frac{13}{n^2} \to 0.$$

Repeated applications of the sum rule, together with the constant limits, therefore yield

$$U_n \longrightarrow 17 + 0 + 0 + 0 = 17, \qquad V_n \longrightarrow 23 + 0 = 23.$$

Moreover, $V_n = 23 + 13/n^2 > 23 > 0$ for every n, and its limit 23 is nonzero. The quotient rule applies and gives

$$\frac{U_n}{V_n} \longrightarrow \frac{17}{23}.$$

Conclusion. The three limits are (a) 1, (b) 1/2, and (c) 17/23. Every use of the quotient theorem has a denominator that is nonzero at every index and has a nonzero finite limit.

9.2 Exercise 9.2: Limits obtained from two given convergent sequences

What we must show. Suppose $x_n \to 3$, $y_n \to 7$, and $y_n \neq 0$ for every n. Find

$$\text{(a)} \quad \lim_{n\to\infty}(x_n + y_n), \qquad \text{(b)} \quad \lim_{n\to\infty} \frac{3y_n - x_n}{y_n^2}.$$

Plan. Use the sum rule in part (a). In part (b), build the numerator from scalar multiples and a sum, build the denominator with the product rule, and then check the quotient rule.

1. Part (a): Add the two limits

The hypotheses give two finite convergent sequences. Theorem 9.3 therefore applies directly:

$$x_n + y_n \longrightarrow 3 + 7 = 10.$$

2. Part (b): Evaluate the numerator

Theorem 9.2 gives

$$3y_n \longrightarrow 3 \cdot 7 = 21, \qquad -x_n \longrightarrow (-1) \cdot 3 = -3.$$

Theorem 9.3 then gives

$$3y_n - x_n = 3y_n + (-x_n) \longrightarrow 21 + (-3) = 18.$$

The numerator here is $3y_n - x_n$; only the denominator contains a square.

3. Part (b): Evaluate the denominator and take the quotient

Since $y_n \to 7$, Theorem 9.4 applied to the two factors y_n gives

$$y_n^2 = y_n\, y_n \longrightarrow 7 \cdot 7 = 49.$$

The assumption $y_n \neq 0$ makes $y_n^2 > 0$ at every index. Thus every denominator is nonzero, and its limit 49 is also nonzero. Theorem 9.6 now gives

$$\frac{3y_n - x_n}{y_n^2} \longrightarrow \frac{18}{49}.$$

Conclusion. The limits are (a) 10 and (b) 18/49.

9.3 Exercise 9.3: A polynomial expression in convergent sequences

What we must show. Assume $a_n \to a \in \mathbb{R}$ and $b_n \to b \in \mathbb{R}$. For

$$s_n = \frac{a_n^3 + 4a_n}{b_n^2 + 1},$$

prove carefully that

$$s_n \longrightarrow \frac{a^3 + 4a}{b^2 + 1}.$$

Plan. Form the powers one multiplication at a time. Then use the scalar and sum rules. The added 1 makes the denominator and its limit positive even when $b_n = 0$ or $b = 0$.

1. Construct the square and cube of the numerator sequence

The two factors in $a_n\,a_n$ both converge to a. The product rule, Theorem 9.4, gives

$$a_n^2 = a_n\,a_n \longrightarrow a \cdot a = a^2.$$

We may now apply the product rule again, this time to a_n^2 and a_n, since both have finite limits:

$$a_n^3 = (a_n^2)a_n \longrightarrow a^2 \cdot a = a^3.$$

Separately, Theorem 9.2 gives $4a_n \to 4a$. The sum rule therefore yields

$$a_n^3 + 4a_n \longrightarrow a^3 + 4a.$$

2. Construct the denominator limit

Apply Theorem 9.4 to $b_n\,b_n$ to obtain

$$b_n^2 \longrightarrow b \cdot b = b^2.$$

The constant sequence 1 converges to 1, so Theorem 9.3 gives

$$b_n^2 + 1 \longrightarrow b^2 + 1.$$

For every real b_n, its square is nonnegative, and thus $b_n^2 + 1 \geq 1 > 0$. The denominator terms are consequently nonzero. Likewise $b^2 + 1 \geq 1 > 0$, so their limit is nonzero.

3. Apply the quotient theorem with all hypotheses established

Step 1 proves convergence of the numerator, Step 2 proves convergence of the denominator, and the positivity checks verify both nonzero requirements. Theorem 9.6 therefore gives the desired limit.

Conclusion. $\lim\limits_{n\to\infty} s_n = \dfrac{a^3 + 4a}{b^2 + 1}$ for all finite real limits a and b, including $b = 0$.

9.4 Exercise 9.4: A nested-square-root recursion

What we must show. For $s_1 = 1$ and $s_{n+1} = \sqrt{s_n + 1}$, list the first four terms and, assuming convergence, prove that the limit is $(1 + \sqrt{5})/2$.

Plan. First check that every radical is defined and every term is at least 1. Under the stated convergence assumption, square the recursion and apply the product and sum limit laws. The lower bound will select the appropriate quadratic root.

1. Verify that the recursion stays defined and positive

The initial value is $s_1 = 1 \geq 1$. If $s_n \geq 1$, then $s_n + 1 \geq 2 > 0$, so its nonnegative real square root exists. Moreover, $\sqrt{s_n + 1} \geq 1$: a nonnegative number less than 1 would have square less than 1, whereas this square is at least 2. Thus $s_{n+1} \geq 1$. Induction shows that all terms are defined and satisfy $s_n \geq 1$.

2. Part (a): Substitute successively four times

The successive terms are

$$\begin{aligned} s_1 &= 1, \\ s_2 &= \sqrt{s_1 + 1} = \sqrt{1+1} = \sqrt{2}, \\ s_3 &= \sqrt{s_2 + 1} = \sqrt{\sqrt{2}+1} = \sqrt{1+\sqrt{2}}, \\ s_4 &= \sqrt{s_3 + 1} = \sqrt{\sqrt{1+\sqrt{2}}+1} = \sqrt{1+\sqrt{1+\sqrt{2}}}. \end{aligned}$$

3. A shift of a convergent sequence has the same limit

We will use this elementary observation for this and the next two recursions. If $u_n \to u \in \mathbb{R}$, then $u_{n+1} \to u$. Indeed, given $\varepsilon > 0$, choose N so that $m > N$ implies $|u_m - u| < \varepsilon$. If $n > N$, then also $n + 1 > N$, so $|u_{n+1} - u| < \varepsilon$. This is exactly the limit definition for the shifted sequence.

4. Part (b): Record the restriction on a finite limit

Assume, as requested, that $s_n \to L \in \mathbb{R}$. Since $s_n \geq 1$ for every n, we have $L \geq 1$. To see the bound directly, if $L < 1$, set $\varepsilon = (1 - L)/2 > 0$. Eventually $|s_n - L| < \varepsilon$, giving

$$s_n < L + \varepsilon = \frac{L+1}{2} < 1,$$

contrary to the bound on the terms. This is also the eventual-bound principle of Exercise 8.9(a).

5. Pass to limits in a squared identity

The defining radical equality implies

$$s_{n+1}^2 = s_n + 1$$

for every n. By Step 3, $s_{n+1} \to L$. Theorem 9.4 applied to $s_{n+1}\, s_{n+1}$ gives $s_{n+1}^2 \to L^2$. Theorem 9.3 gives $s_n + 1 \to L + 1$. These two expressions are equal term by term, so uniqueness of finite limits, proved at the end of Section 7, gives

$$L^2 = L + 1.$$

Only the product and sum laws were used; a square-root limit rule is not needed.

6. Solve the quadratic equation and select its permitted root

Rearrange and complete the square:

$$L^2 - L - 1 = 0, \qquad 4L^2 - 4L = 4, \qquad (2L-1)^2 = 5.$$

Consequently $2L - 1 = \sqrt{5}$ or $2L - 1 = -\sqrt{5}$, which gives

$$L = \frac{1+\sqrt{5}}{2} \quad \text{or} \quad L = \frac{1-\sqrt{5}}{2}.$$

We have $\sqrt{5} > 1$, since the opposite nonnegative inequality would imply $5 \leq 1$ after squaring. Thus $(1 - \sqrt{5})/2 < 0$, contradicting $L \geq 1$. The only remaining root is $(1 + \sqrt{5})/2$.

Conclusion. The first four terms are 1, $\sqrt{2}$, $\sqrt{1+\sqrt{2}}$, and $\sqrt{1+\sqrt{1+\sqrt{2}}}$. Under the given convergence assumption, $\lim s_n = (1 + \sqrt{5})/2$.

9.5 Exercise 9.5: A positive rational recursion

What we must show. Let $t_1 = 1$ and

$$t_{n+1} = \frac{t_n^2 + 2}{2t_n} \qquad (n \geq 1).$$

Assuming that (t_n) converges to a finite real number, find its limit.

Plan. Prove positivity so that every term is defined, and strengthen it to the uniform lower bound $t_n \geq 1$. The lower bound makes the assumed limit nonzero before the quotient theorem is applied.

1. Check that the recursion is defined at every index

We start with $t_1 = 1 > 0$. If $t_n > 0$, then its denominator $2t_n$ is positive and its numerator $t_n^2 + 2$ is positive. The quotient is therefore a well-defined positive real number. Induction proves $t_n > 0$ for all n, and in particular no denominator in the recursion vanishes.

2. Obtain a positive uniform lower bound

For every n, use the positive denominator to compute

$$\begin{aligned} t_{n+1} - 1 &= \frac{t_n^2 + 2}{2t_n} - \frac{2t_n}{2t_n} \\ &= \frac{t_n^2 - 2t_n + 2}{2t_n} \\ &= \frac{(t_n - 1)^2 + 1}{2t_n} > 0. \end{aligned}$$

The last numerator is positive because $(t_n - 1)^2 \geq 0$. Thus every term after t_1 is greater than 1, and $t_1 = 1$. It follows that $t_n \geq 1$ for every n.

3. The convergence assumption gives a nonzero limit

Suppose $t_n \to L \in \mathbb{R}$. The bound $t_n \geq 1$ passes to the limit by Exercise 8.9(a), so $L \geq 1 > 0$. Equivalently, if $L < 1$, the positive tolerance $(1 - L)/2$ would force eventual terms to be less than $L + (1 - L)/2 = (L + 1)/2 < 1$, a contradiction. In particular, $2L \neq 0$.

4. Apply the finite limit laws to the recursion

The product rule, Theorem 9.4, gives

$$t_n^2 = t_n \, t_n \longrightarrow L^2.$$

The constant sequence 2 converges to 2, and hence the sum rule gives $t_n^2 + 2 \to L^2 + 2$. The scalar rule gives $2t_n \to 2L$. The terms $2t_n$ are nonzero by Step 1, and their limit $2L$ is nonzero by Step 3. Thus Theorem 9.6 applies:

$$\frac{t_n^2 + 2}{2t_n} \longrightarrow \frac{L^2 + 2}{2L}.$$

The shift observation in Exercise 9.4 also gives $t_{n+1} \to L$. The recursion identifies these two sequences, so uniqueness of finite limits gives

$$L = \frac{L^2 + 2}{2L}.$$

5. Solve and select the positive root

Since $2L \neq 0$, multiplying the last equality by $2L$ is valid:

$$2L^2 = L^2 + 2, \qquad L^2 = 2.$$

Thus $L = \sqrt{2}$ or $L = -\sqrt{2}$. The root $-\sqrt{2}$ is negative and violates $L \geq 1$. Consequently the assumed finite limit is $\sqrt{2}$.

Conclusion. Under the convergence assumption, $\lim_{n\to\infty} t_n = \sqrt{2}$. The nonzero denominator limit was established before the quotient theorem was used.

9.6 Exercise 9.6: A necessary finite-limit equation and an actual infinite limit

What we must show. For $x_1 = 1$ and $x_{n+1} = 3x_n^2$, address all three questions: (a) determine the possibilities for a finite real limit; (b) decide whether the actual sequence has a limit; (c) explain why the conclusions are consistent.

Plan. A hypothetical finite limit must satisfy the recursion's fixed-point equation. Separately, use an induction estimate for the actual terms. Distinguish finite convergence from the extended infinite-limit convention in Definition 9.8.

1. Part (a): Assume a finite limit before applying the product rule

Suppose $x_n \to a \in \mathbb{R}$. The shift observation proved in Exercise 9.4 gives $x_{n+1} \to a$. Theorem 9.4 gives $x_n^2 \to a^2$, and Theorem 9.2 then gives $3x_n^2 \to 3a^2$. Since $x_{n+1} = 3x_n^2$ for every n, uniqueness of finite limits implies

$$a = 3a^2.$$

Subtracting a and factoring gives

$$0 = 3a^2 - a = a(3a - 1).$$

The zero-product property in $\mathbb{R}$ says that one of the two factors must be zero. Thus $a = 0$ or $3a - 1 = 0$; the second equation gives $a = 1/3$ after division by $3 \neq 0$.

This proves only a necessary condition: *if the sequence has a finite real limit, that limit must be* 0 *or* $1/3$. The computation does not prove that such a finite limit exists.

2. Part (b): Bound the actual terms below by their indices

We prove $x_n \geq n$ for every $n \in \mathbb{N}$. For $n = 1$, the equality $x_1 = 1$ supplies the base case. Suppose $x_n \geq n$ for some $n \geq 1$. In particular $x_n \geq 1$, so $x_n^2 - x_n = x_n(x_n - 1) \geq 0$. It follows that

$$x_{n+1} = 3x_n^2 \geq 3x_n \geq 3n \geq n + 1.$$

The final inequality holds because $3n - (n + 1) = 2n - 1 \geq 1 > 0$ for $n \geq 1$. This completes the induction.

Now let $M > 0$ and take the threshold $N = M$. For every integer $n > N$, the bound gives

$$x_n \geq n > M.$$

This is exactly Definition 9.8 for $x_n \to +\infty$. Thus the actual sequence has the extended limit $+\infty$.

It cannot have a finite real limit. For a direct check, if $x_n \to a \in \mathbb{R}$, then eventually $|x_n - a| < 1$, which implies $x_n < a + 1$. Choose an integer n larger than both that eventual threshold and $a + 1$, using the Archimedean property. The bound $x_n \geq n$ would instead give $x_n > a + 1$, a contradiction. This is also consistent with Theorem 9.1, which says every finitely convergent sequence is bounded.

3. Part (c): Explain the two meanings of the limit statement

Part (a) began with the hypothesis of a *finite real* limit a. Its use of the finite product and scalar limit laws is valid under that hypothesis. Part (b) shows that the hypothesis fails for this initial value: the terms grow without bound. Consequently the necessary alternatives in part (a) are not attained by this sequence.

Under the convention introduced after Definition 9.8, saying that "the limit exists" also allows divergence to $+\infty$ or $-\infty$. In that stated convention the answer to part (b) is yes, with limit $+\infty$; the sequence does not converge to a finite real number. Theorems 9.2–9.6 were proved for finite real limits, so their product calculation in part (a) cannot be applied with $a = +\infty$. The extended infinite-limit conclusion therefore supplies no contradiction to the finite necessary condition.

Conclusion. (a) Any hypothetical finite real limit is 0 or 1/3. (b) The actual sequence satisfies $\lim x_n = +\infty$ in the sense of Definition 9.8, and has no finite real limit. (c) A necessary condition for finite convergence does not assert its existence, and finite product laws cannot be applied at an infinite limit.

9.7 Exercise 9.7: Finish the estimate for the root of the index

What we must show. Complete the argument that

$$s_n = n^{1/n} - 1 \longrightarrow 0,$$

and hence finish the proof of Theorem 9.7(c), $n^{1/n} \to 1$.

Plan. Use the nonnegative binomial terms in $(1 + s_n)^n$ to obtain an explicit bound on s_n. Then choose a threshold directly in terms of the prescribed positive tolerance.

1. Verify the sign and the defining power identity

Let $u_n = n^{1/n}$ denote the positive real nth root of n. Then $u_n^n = n \geq 1$. If $u_n < 1$, the positivity $0 < u_n < 1$ would give $u_n^n < 1$, a contradiction. Hence $u_n \geq 1$ and

$$s_n = u_n - 1 \geq 0, \qquad (1 + s_n)^n = u_n^n = n.$$

For $n = 1$, this gives $s_1 = 1 - 1 = 0$.

2. Derive the binomial bound for $n \geq 2$

For an integer $n \geq 2$, the binomial theorem gives

$$n = (1 + s_n)^n = 1 + ns_n + \frac{n(n-1)}{2} s_n^2 + \sum_{j=3}^{n} \binom{n}{j} s_n^j.$$

When $n = 2$, the final sum is empty and equals 0. All terms of that sum are nonnegative, since $s_n \geq 0$ and all the binomial coefficients are nonnegative. Also $ns_n \geq 0$, and the constant term 1 is strictly positive. Therefore

$$n > \frac{n(n-1)}{2} s_n^2.$$

Dividing by the positive number n gives $1 > (n-1)s_n^2/2$. Multiplying by $2/(n-1) > 0$ yields

$$s_n^2 < \frac{2}{n-1}.$$

Both s_n and the square root of the right-hand side are nonnegative, so the order of their squares gives

$$0 \le s_n < \sqrt{\frac{2}{n-1}} \qquad (n \ge 2).$$

3. Specify the threshold and verify the limit definition

Let $\varepsilon > 0$ be arbitrary and set

$$N = \max\left\{2,\ 1 + \frac{2}{\varepsilon^2}\right\}.$$

A real threshold is permitted by the definition of convergence; an integer larger than it may also be chosen by the Archimedean property. If $n > N$, then $n \ge 2$ and

$$n - 1 > \frac{2}{\varepsilon^2} > 0.$$

Taking reciprocals of positive quantities and multiplying by 2 gives $2/(n-1) < \varepsilon^2$. As $\varepsilon > 0$, this implies

$$\sqrt{\frac{2}{n-1}} < \varepsilon.$$

Using the bound from Step 2, we conclude

$$|s_n - 0| = s_n < \sqrt{\frac{2}{n-1}} < \varepsilon \qquad (n > N).$$

This is the required proof that $s_n \to 0$.

4. Return to the root sequence

The constant sequence 1 converges to 1, and $s_n \to 0$ was proved without assuming a limit for $n^{1/n}$. The sum rule, Theorem 9.3, therefore gives

$$n^{1/n} = 1 + s_n \longrightarrow 1 + 0 = 1.$$

Conclusion. The explicit estimate $0 \le n^{1/n} - 1 < \sqrt{2/(n-1)}$ for $n \ge 2$, together with $N = \max\{2, 1 + 2/\varepsilon^2\}$, proves $n^{1/n} - 1 \to 0$ and hence $n^{1/n} \to 1$.

9.8 Exercise 9.8: Finite and infinite possibilities for five limits

What we must show. Determine each of the five requested limits, using the convention introduced in Definition 9.8: an existing limit may be a real number, $+\infty$, or $-\infty$. The answer "NOT EXIST" therefore means that none of these three possibilities works.

Plan. Use the defining threshold tests. A sequence has limit $+\infty$ if, for every $M > 0$, all sufficiently late terms exceed M. It has limit $-\infty$ if, for every $M < 0$, all sufficiently late terms are below M.

Neither definition describes finite convergence, and the finite-limit arithmetic laws cannot simply be applied to expressions involving infinity.

(a) The sequence n^3.

Let $M > 0$ be arbitrary. Set $N = \max\{1, M\}$. If $n > N$, then $n \geq 1$ and

$$n^3 = n \cdot n^2 \geq n > M.$$

Thus every prescribed positive threshold is exceeded by all sufficiently late terms, exactly as Definition 9.8 requires.

Conclusion. In part (a), $\lim_{n\to\infty} n^3 = +\infty$.

(b) The sequence $-n^3$.

Let $M < 0$. The number $-M$ is positive. Choose $N = \max\{1, -M\}$. If $n > N$, the estimate used in (a) gives $n^3 \geq n > -M$. Multiplication by -1 reverses the inequality, giving $-n^3 < M$. This is the negative-infinity definition.

Conclusion. In part (b), $\lim_{n\to\infty}(-n^3) = -\infty$.

(c) The sequence $(-n)^n$.

At an even index $n = 2m$ the value is $(2m)^{2m} > 0$; at an odd index $n = 2m+1$ it is $-(2m+1)^{2m+1} < 0$. There are even and odd indices beyond every threshold: choose a positive integer $m > N$ to make both $2m$ and $2m + 1$ exceed a prescribed N.

The limit cannot be $+\infty$, because the test with $M = 1$ would require all sufficiently late terms to be greater than 1, whereas arbitrarily late odd terms are negative. It cannot be $-\infty$, because the test with $M = -1$ would require all sufficiently late terms to be less than -1, whereas arbitrarily late even terms are positive.

It also cannot be a finite real number L. If it converged to L, the definition with $\varepsilon = 1$ would give a threshold N_1 such that

$$n > N_1 \quad \Longrightarrow \quad |(-n)^n - L| < 1.$$

Choose an even positive integer $n > N_1$ with $n > \max\{1, L + 1\}$. Such an index exists by taking $n = 2m$ with m sufficiently large. Since $n \geq 1$, $n^n \geq n$, so at this even index

$$(-n)^n = n^n \geq n > L + 1.$$

Consequently $|(-n)^n - L| = n^n - L > 1$, contradicting the required error bound. All three possible types of limit have now been excluded.

Conclusion. In part (c), the limit does NOT EXIST, even in the extended sense of Definition 9.8.

(d) The sequence $(1.01)^n$.

Write $1.01 = 1 + 1/100$. The binomial theorem, or Bernoulli's inequality, gives

$$\left(1 + \frac{1}{100}\right)^n \geq 1 + \frac{n}{100}.$$

For completeness, the inequality $(1 + x)^n \geq 1 + nx$ for $x \geq 0$ follows by induction: the case $n = 1$ is equality, and from the inequality at n one obtains

$$(1 + x)^{n+1} \geq (1 + nx)(1 + x) = 1 + (n + 1)x + nx^2 \geq 1 + (n + 1)x.$$

Now let $M > 0$ and set $N = 100M$. If $n > N$, then $n/100 > M$, and hence

$$(1.01)^n \geq 1 + \frac{n}{100} > M.$$

This proves divergence to positive infinity by the definition.

Conclusion. In part (d), $\lim_{n\to\infty} (1.01)^n = +\infty$.

(e) The sequence n^n.

For $n \geq 1$, the product of n factors all equal to n satisfies $n^n \geq n$. Let $M > 0$ and set $N = \max\{1, M\}$. If $n > N$, then $n^n \geq n > M$. The definition therefore applies.

Conclusion. In part (e), $\lim_{n\to\infty} n^n = +\infty$.

9.9 Exercise 9.9: Order preserved by existing extended limits

What we must show. Assume there is a threshold N_0 such that $s_n \leq t_n$ whenever $n > N_0$. Prove the three assertions, including every possible finite or infinite case in part (c).

Plan. Combine the threshold at which the comparison becomes valid with the threshold supplied by the relevant limit. In the finite case, two limits in the wrong order would force the late terms to have the wrong order as well.

(a) From $s_n \to +\infty$ to $t_n \to +\infty$.

Fix $M > 0$. Since $s_n \to +\infty$, some N_1 satisfies $s_n > M$ for all $n > N_1$. Set $N = \max\{N_0, N_1\}$. For every $n > N$, both the comparison and this threshold test hold, so

$$t_n \geq s_n > M.$$

This holds for the arbitrary $M > 0$, proving $t_n \to +\infty$.

(b) From $t_n \to -\infty$ to $s_n \to -\infty$.

Fix $M < 0$. Since $t_n \to -\infty$, some N_1 satisfies $t_n < M$ for all $n > N_1$. With $N = \max\{N_0, N_1\}$, every $n > N$ gives

$$s_n \leq t_n < M.$$

This is the definition of $s_n \to -\infty$.

(c) Account for the infinite limit cases.

Write $S = \lim s_n$ and $T = \lim t_n$, which exist by the assumption in this part. Their possible values lie in $\mathbb{R} \cup \{-\infty, +\infty\}$, with the usual extended order. The finite and infinite alternatives cannot overlap. Indeed, convergence to a real c puts the late terms between $c - 1$ and $c + 1$. A simultaneous limit $+\infty$ would put them above $\max\{1, c + 2\}$, contradicting the upper restriction; a simultaneous limit $-\infty$ would put them below $\min\{-1, c - 2\}$, contradicting the lower restriction. Limits of both signs of infinity are incompatible as well, since they would eventually require a term to be both greater than 1 and less than -1. The following cases exhaust all possibilities involving infinity:

1. If $S = -\infty$, then $S \leq T$ for every possible T.

2. If $T = +\infty$, then $S \leq T$ for every possible S.

3. If $S = +\infty$, part (a) forces $T = +\infty$, so $S = T$ and in particular $S \le T$.

4. If $T = -\infty$, part (b) forces $S = -\infty$, so again $S = T$.

After these cases, both S and T must be finite real numbers.

(c) Prove the finite case by contradiction.

Suppose, contrary to the desired inequality, that $S > T$. Set $\delta = (S-T)/3 > 0$. By finite convergence there are thresholds N_1, N_2 such that

$$n > N_1 \Longrightarrow |s_n - S| < \delta, \qquad n > N_2 \Longrightarrow |t_n - T| < \delta.$$

Thus $s_n > S - \delta$ and $t_n < T + \delta$ at the corresponding late indices. But

$$(S-\delta) - (T+\delta) = (S-T) - 2\delta = (S-T) - \frac{2(S-T)}{3} = \frac{S-T}{3} > 0.$$

For $n > \max\{N_0, N_1, N_2\}$ this gives

$$s_n > S - \delta > T + \delta > t_n,$$

contradicting $s_n \le t_n$. Hence $S \le T$ in the finite case as well.

Conclusion. The eventual order implies both infinite-limit comparisons in (a) and (b), and $\lim s_n \le \lim t_n$ whenever both limits exist in the extended sense.

9.10 Exercise 9.10: Scaling and reversing an infinite limit

What we must show. Prove the effects of multiplying a sequence with limit $+\infty$ by a positive constant or a negative constant, and prove both directions of the sign-reversal equivalence.

Plan. Choose a threshold for s_n by dividing the target threshold by the multiplying constant. A negative constant reverses the inequality, so the signs must be recorded.

(a) Multiplication by $k > 0$.

Assume $s_n \to +\infty$ and fix $k > 0$. To prove $ks_n \to +\infty$, let $M > 0$. Then $M/k > 0$, so the assumed limit supplies N with

$$n > N \quad \Longrightarrow \quad s_n > \frac{M}{k}.$$

Multiplying by $k > 0$ preserves the direction and gives $ks_n > M$. This verifies the target definition.

In fact this implication is reversible. If $ks_n \to +\infty$ and $M > 0$, use the positive threshold kM for ks_n. Eventually $ks_n > kM$, and division by $k > 0$ gives $s_n > M$. Thus positive multiplication preserves the $+\infty$ limit in both directions.

(b) Prove $s_n \to +\infty \Longrightarrow -s_n \to -\infty$.

Let $M < 0$ be an arbitrary threshold for $-s_n$. Because $-M > 0$, the assumption $s_n \to +\infty$ supplies N such that $n > N$ implies $s_n > -M$. Multiplication by -1 reverses the inequality, giving $-s_n < M$. Thus $-s_n \to -\infty$.

(b) Prove the reverse implication.

Assume $-s_n \to -\infty$ and let $M > 0$. The number $-M$ is negative, so some N satisfies $n > N \implies -s_n < -M$. Multiplication by -1 reverses this inequality and gives $s_n > M$. Consequently $s_n \to +\infty$, completing the equivalence.

(c) Multiplication by $k < 0$.

Assume $s_n \to +\infty$ and $k < 0$. Let $M < 0$ be an arbitrary target threshold for ks_n. The quotient M/k is positive, since both numerator and denominator are negative. Hence the assumed limit supplies N such that $n > N$ implies $s_n > M/k$. Multiplying by $k < 0$ reverses the comparison:

$$n > N \quad \implies \quad ks_n < M.$$

This proves $ks_n \to -\infty$.

The corresponding reverse implication also holds. If $ks_n \to -\infty$, fix $M > 0$ and use the negative threshold kM. Eventually $ks_n < kM$, and division by $k < 0$ reverses the inequality, yielding $s_n > M$. Therefore $s_n \to +\infty$.

Conclusion. Positive scaling preserves a positive-infinity limit; negative scaling changes it to negative infinity. In particular, $s_n \to +\infty$ if and only if $-s_n \to -\infty$, with each direction justified by its own threshold test.

9.11 Exercise 9.11: Adding a sequence that is bounded below

What we must show. Prove the sum has limit $+\infty$ in each of the three stated situations. Distinguish a lower bound valid for all terms from a lower bound valid only after an initial segment.

Plan. Once $t_n \geq m$ holds in the tail, it suffices to make $s_n > M - m$. If this number is not positive, replace it by a larger positive threshold before invoking Definition 9.8.

(a) Use the stated global lower bound.

Assume $s_n \to +\infty$ and $\inf\{t_n : n \in \mathbb{N}\} > -\infty$. The set is nonempty and consists of real numbers; its infimum is at most the finite member t_1. Thus this hypothesis provides a real lower bound m, for example the real infimum itself, such that $t_n \geq m$ for every n.

Let $M > 0$ be arbitrary, and define

$$B = \max\{1, M - m\} > 0.$$

Since $s_n \to +\infty$, there is N such that $n > N$ implies $s_n > B$. For those indices,

$$s_n + t_n \geq s_n + m > B + m \geq (M - m) + m = M.$$

The first inequality uses the global lower bound; the last uses $B \geq M - m$. This proves the positive-infinity limit.

(b) Derive an eventual lower bound in every allowed limit case.

Assume $s_n \to +\infty$ and $T = \lim t_n > -\infty$. The existing limit T is either a real number or $+\infty$.

- If $T \in \mathbb{R}$, apply finite convergence with $\varepsilon = 1$. For some N_1, every $n > N_1$ has $|t_n - T| < 1$, implying $t_n > T - 1$. Thus we may take $m = T - 1$ as an eventual lower bound.

- If $T = +\infty$, use the positive threshold 1 in Definition 9.8. For some N_1, every $n > N_1$ has $t_n > 1$. Thus we may take $m = 1$.

In either case there are a real m and a threshold N_1 such that $t_n \geq m$ whenever $n > N_1$. This is a tail bound; no assertion about the earlier terms is needed.

Now fix $M > 0$. Set $B = \max\{1, M - m\} > 0$, and use $s_n \to +\infty$ to choose N_2 such that $s_n > B$ for $n > N_2$. For $n > \max\{N_1, N_2\}$, both estimates hold and

$$s_n + t_n \geq s_n + m > B + m \geq M.$$

This proves the desired limit using the eventual bound directly, without applying a global hypothesis to an unexamined prefix.

(c) A bounded sequence is bounded below.

If (t_n) is bounded, there is a real $C \geq 0$ with $|t_n| \leq C$ for every n. Therefore $t_n \geq -C$ for every n, a global lower bound, and part (a) applies. One can also see the threshold explicitly: for $M > 0$, choose N such that $s_n > M + C$ whenever $n > N$; this is a valid positive threshold because $M + C > 0$. Then

$$s_n + t_n \geq s_n - C > M + C - C = M.$$

Conclusion. In (a), (b), and (c), the sum $s_n + t_n$ tends to $+\infty$. An eventual lower bound on the added sequence already suffices, as the proof of (b) shows.

9.12 Exercise 9.12: The ratio test for the size of a sequence

What we must show. Suppose $s_n \neq 0$ for every n, and the limit

$$L = \lim_{n\to\infty} r_n, \qquad r_n = \left|\frac{s_{n+1}}{s_n}\right|,$$

exists. Prove $s_n \to 0$ if $L < 1$, and $|s_n| \to +\infty$ if $L > 1$, including the case $L = +\infty$.

Plan. A ratio eventually less than a fixed $q < 1$ gives a geometric bound. For ratios tending to a value larger than 1, apply that first result to the reciprocal magnitudes.

First locate the possible values of L.

Every r_n is positive, since its numerator and denominator are nonzero in absolute value. Therefore its existing limit can only be a nonnegative real number or $+\infty$. For example, if a finite limit were $L < 0$, convergence with $\varepsilon = -L/2 > 0$ would eventually give $r_n < L + (-L/2) = L/2 < 0$, a contradiction. A limit $-\infty$ would eventually give $r_n < -1$ by its definition, also a contradiction. This verifies the claimed restriction on L.

(a) Choose a single ratio bound valid at every sufficiently late index.

Suppose $L < 1$. Then $0 \leq L < 1$, so

$$q = \frac{L+1}{2} \quad \text{satisfies} \quad 0 < q < 1 \quad \text{and} \quad L < q.$$

By convergence of r_n to the finite number L, some threshold N_0 gives $|r_n - L| < q - L$ for all $n > N_0$. In particular $r_n < q$ at all those indices. Choose an integer $N \in \mathbb{N}$ with $N > N_0$. Then every integer $n \geq N$ has $n > N_0$, so

$$|s_{n+1}| = r_n \, |s_n| < q \, |s_n| \qquad (n \geq N).$$

The integer choice makes the starting index and the validity of every subsequent ratio inequality explicit.

(a) Prove the geometric estimate by induction.

For $n = N + 1$, the preceding inequality gives

$$|s_{N+1}| < q \, |s_N| = q^{(N+1)-N} \, |s_N| \, .$$

Assume for some integer $n > N$ that $|s_n| < q^{n-N} \, |s_N|$. Since the ratio estimate also applies at this n, multiplication by $q > 0$ gives

$$|s_{n+1}| < q \, |s_n| < q \, q^{n-N} \, |s_N| = q^{(n+1)-N} \, |s_N| \, .$$

Induction proves

$$|s_n| < q^{n-N} \, |s_N| \qquad (n > N).$$

Since $0 < q < 1$, Theorem 9.7(b) gives $q^m \to 0$ as $m \to \infty$. To spell out the use of this fact, let $\varepsilon > 0$. The number $C = |s_N|$ is positive, because $s_N \neq 0$. Choose a threshold K such that $m > K$ implies $q^m < \varepsilon / C$. If $n > N + \max\{K, 0\}$, then $m = n - N > K$ and $n > N$, so

$$|s_n - 0| < q^{n-N} C < \frac{\varepsilon}{C} C = \varepsilon.$$

This is precisely finite convergence to 0.

(b) Form the reciprocal sequence and compute its ratio.

Suppose $L > 1$, and define

$$t_n = \frac{1}{|s_n|} > 0.$$

It is defined and nonzero for every n. Its absolute ratio is

$$\begin{aligned} \left| \frac{t_{n+1}}{t_n} \right| &= \frac{1/\,|s_{n+1}|}{1/\,|s_n|} \\ &= \frac{|s_n|}{|s_{n+1}|} = \frac{1}{r_n}. \end{aligned}$$

There are two distinct cases for the ratio limit:

- If $1 < L < +\infty$, the finite reciprocal law applies to $r_n \to L$, since every $r_n \neq 0$ and $L \neq 0$. It gives $1/r_n \to 1/L$, with $0 < 1/L < 1$.
- If $L = +\infty$, every r_n is positive, so Theorem 9.10 gives $1/r_n \to 0$. This conclusion uses the infinite-limit reciprocal theorem, not the finite quotient law with an infinite denominator.

In either case, the absolute ratio for t_n has a finite limit strictly less than 1. Part (a) therefore proves $t_n \to 0$. Finally apply Theorem 9.10 to the positive sequence $|s_n|$: its reciprocal is $t_n \to 0$, so

$$|s_n| \longrightarrow +\infty.$$

Conclusion. If $L < 1$, the sequence itself converges to 0. If $L > 1$, including $L = +\infty$, its magnitudes tend to $+\infty$. No conclusion for the boundary value $L = 1$ is asserted by either part.

9.13 Exercise 9.13: All real bases in a geometric sequence

What we must show. Prove the complete classification of a^n as the fixed real base a varies, including $a = 0$, $a = 1$, $a = -1$, and both intervals of negative bases.

Plan. Use Exercise 9.12 where the terms are nonzero. For a negative base with magnitude at least 1, check that no finite limit and neither infinite limit can accommodate the alternating signs.

Case 1: $a = 0$, and then $0 < |a| < 1$.

If $a = 0$, every term a^n is 0 because $n \geq 1$. Thus the sequence converges to 0 directly. If $0 < |a| < 1$, every term is nonzero and

$$\left|\frac{a^{n+1}}{a^n}\right| = |a| < 1.$$

This constant ratio has limit $|a|$, so Exercise 9.12(a) proves $a^n \to 0$. The two cases together cover all $|a| < 1$, including negative a of magnitude less than 1.

Case 2: $a = 1$.

Every term equals $1^n = 1$. Thus for every $\varepsilon > 0$, $|a^n - 1| = 0 < \varepsilon$ at every index, giving limit 1.

Case 3: $a > 1$.

The terms a^n are positive and nonzero. Their absolute ratio is the constant $a > 1$. Exercise 9.12(b) gives $|a^n| \to +\infty$. Since $a^n > 0$, $|a^n| = a^n$, so the sequence itself tends to $+\infty$.

Case 4: $a = -1$.

The terms are 1 at even indices and -1 at odd indices. If there were a finite limit L, convergence with $\varepsilon = 1/2$ would put every sufficiently late term within $1/2$ of L. Choose a late even index p and a late odd index q. The triangle inequality would give

$$2 = |1 - (-1)| \leq |a^p - L| + |L - a^q| < \frac{1}{2} + \frac{1}{2} = 1,$$

a contradiction. A limit $+\infty$ fails the threshold $M = 1$ because late odd terms equal -1. A limit $-\infty$ fails the threshold $M = -1$ because late even terms equal 1. Thus no extended limit exists.

Case 5: $a < -1$.

Set $b = |a| = -a > 1$. By Case 3, $b^n \to +\infty$. The terms satisfy

$$a^n = (-1)^n b^n, \qquad |a^n| = b^n.$$

Suppose first that a^n had a finite limit L. Convergence with $\varepsilon = 1$ would give, eventually,

$$|a^n| \leq |a^n - L| + |L| < 1 + |L|\,.$$

But $b^n \to +\infty$ means that eventually $b^n > 1 + |L|$, contradicting $|a^n| = b^n$. Thus there is no finite limit.

There is also no limit $+\infty$, since for every threshold index there is a later odd index with $a^n < 0$, defeating the positive threshold $M = 1$. There is no limit $-\infty$, since there is always a later even index with $a^n > 0$, defeating the negative threshold $M = -1$. The required even and odd indices can be taken as $2m$ and $2m + 1$, where m is any positive integer larger than the proposed threshold index.

Conclusion. For every real a,

$$\lim_{n\to\infty} a^n = \begin{cases} 0, & |a| < 1, \\ 1, & a = 1, \\ +\infty, & a > 1, \\ \text{does not exist}, & a \le -1. \end{cases}$$

The last line excludes finite limits and both infinite limits.

9.14 Exercise 9.14: Geometric growth compared with a real power

What we must show. Fix $p > 0$ and determine the limit of a^n/n^p for every real a, using Exercise 9.12 where its hypotheses apply. In particular, include both boundary bases $a = 1$ and $a = -1$.

Plan. The absolute ratio is $|a|\ (n/(n+1))^p$. Prove its second factor tends to 1, then distinguish ratios below 1, above 1, and the boundary cases where the ratio test does not decide the answer.

1. State the real-power facts and justify the needed limit.

As in Theorem 9.7(a), positive bases with real exponents use the usual real-power algebra and order: products and quotients obey the exponent rules; and $r^v \le r^u$ whenever $0 < r < 1$ and $0 < u < v$. These are the elementary real-power properties already implicit in the statement of Theorem 9.7(a). We do not assume a continuity theorem for real powers from a later section.

Set $r_n = n/(n+1)$, so $0 < r_n < 1$. Choose an integer $m > p$, possible by the Archimedean property. Then

$$r_n^m \le r_n^p \le 1, \qquad 0 \le 1 - r_n^p \le 1 - r_n^m.$$

For the integer exponent m, finite multiplication gives the factorization

$$1 - r_n^m = (1 - r_n)(1 + r_n + \cdots + r_n^{m-1}).$$

Every one of the m terms in the parenthesized sum is at most 1, and $1 - r_n = 1/(n+1)$. Therefore

$$0 \le 1 - r_n^p \le 1 - r_n^m \le \frac{m}{n+1}.$$

The final expression tends to 0: for $\varepsilon > 0$, $n > m/\varepsilon$ makes $m/(n+1) < \varepsilon$. This direct error estimate proves

$$\left(\frac{n}{n+1}\right)^p \longrightarrow 1$$

for the possibly nonintegral exponent $p > 0$.

2. Compute the ratio when $a \neq 0$.

Let $u_n = a^n/n^p$. Since $n > 0$, its denominator is positive. If $a \neq 0$, all u_n are nonzero, and the real-power quotient rule yields

$$\left|\frac{u_{n+1}}{u_n}\right| = \left|\frac{a^{n+1}}{(n+1)^p} \cdot \frac{n^p}{a^n}\right| = |a|\ \frac{n^p}{(n+1)^p}$$

$$= |a| \left(\frac{n}{n+1} \right)^p.$$

Step 1 and the finite scalar-multiple limit theorem show that this ratio has limit $|a|$.

3. Treat $a = 0$ and the cases $0 < |a| < 1$.

For $a = 0$, every numerator is 0 and every denominator is positive, so $u_n = 0$ and the limit is 0. For $0 < |a| < 1$, Step 2 gives an absolute ratio limit less than 1, so Exercise 9.12(a) applies and gives $u_n \to 0$.

4. Treat the two boundary cases $|a| = 1$.

If $a = 1$ or $a = -1$, then $|a^n| = 1$, and hence

$$|u_n - 0| = \frac{1}{n^p}.$$

Theorem 9.7(a), whose hypothesis is precisely $p > 0$, says $1/n^p \to 0$. Thus $u_n \to 0$ in both boundary cases. The ratio limit here is 1; the conclusion comes from the direct error bound rather than from an inapplicable strict case of Exercise 9.12.

5. Treat $a > 1$.

The absolute ratio limit is $a > 1$, so Exercise 9.12(b) gives $|u_n| \to +\infty$. Since $a^n > 0$ and $n^p > 0$, the terms themselves are positive: $u_n = |u_n|$. Consequently $u_n \to +\infty$.

6. Treat $a < -1$ and exclude all three types of limit.

Exercise 9.12(b) gives $|u_n| \to +\infty$, since the absolute ratio limit is $|a| > 1$. If u_n converged to a finite L, then eventually

$$|u_n| \leq |u_n - L| + |L| < 1 + |L|$$

by the error bound $|u_n - L| < 1$. This contradicts $|u_n| \to +\infty$. At even indices $u_n > 0$, while at odd indices $u_n < 0$. Every tail has an odd index, so the positive-infinity test with $M = 1$ fails. Every tail also has an even index, so the negative-infinity test with $M = -1$ fails. Thus no extended limit exists in this last case.

Conclusion. For $p > 0$ and real a,

$$\lim_{n\to\infty} \frac{a^n}{n^p} = \begin{cases} 0, & |a| \leq 1, \\ +\infty, & a > 1, \\ \text{does not exist}, & a < -1. \end{cases}$$

In particular, the base $a = -1$ now gives limit 0, unlike the unscaled sequence in Exercise 9.13.

9.15 Exercise 9.15: Every fixed geometric sequence is dominated by a factorial

What we must show. Prove $a^n/n! \to 0$ for every $a \in \mathbb{R}$, including zero and negative values.

Plan. After the zero-base case is separated, the absolute ratio is $|a|/(n+1)$, which tends to 0. This gives the strict ratio condition from Exercise 9.12(a).

1. Handle $a = 0$ without dividing by a zero term.

When $a = 0$ and $n \geq 1$, $a^n = 0$. Since $n! > 0$, every term is 0, and the limit is 0.

2. Compute the absolute ratio for $a \neq 0$.

Fix a nonzero real a and write $s_n = a^n/n!$. Then every s_n is nonzero. By the factorial identity $(n+1)! = (n+1)n!$,

$$\begin{aligned}\left|\frac{s_{n+1}}{s_n}\right| &= \left|\frac{a^{n+1}}{(n+1)!} \cdot \frac{n!}{a^n}\right| \\ &= \left|\frac{a\, n!}{(n+1)n!}\right| \\ &= \frac{|a|}{n+1}.\end{aligned}$$

For $\varepsilon > 0$, choosing $N > |a|/\varepsilon$ makes $n > N$ imply $|a|/(n+1) < \varepsilon$. Thus the ratio limit is $L = 0 < 1$. Exercise 9.12(a) now proves $s_n \to 0$. Its hypothesis is about absolute ratios, so this argument applies to negative a as well as positive a.

Conclusion. For every real a, $\lim_{n\to\infty} a^n/n! = 0$.

9.16 Exercise 9.16: Three applications of the infinite-limit results

What we must show. Establish positive-infinity limits for all three given expressions, using the proved infinite-limit theorems or Exercises 9.9–9.15 with their hypotheses checked.

Plan. In (a), isolate a factor with limit $+\infty$ and another with a positive finite limit. In (b) and (c), use Exercise 9.14 for the growing term and then control the added term from below.

(a) The expression $(n^4 + 8n)/(n^2 + 9)$.

Every denominator $n^2 + 9$ is positive. Factor the expression exactly as

$$\frac{n^4 + 8n}{n^2 + 9} = n^2\,\frac{1 + 8/n^3}{1 + 9/n^2}.$$

Indeed, multiplying the numerator and denominator of the right-hand fraction by n^2 gives

$$n^2\,\frac{n^2 + 8/n}{n^2 + 9} = \frac{n^4 + 8n}{n^2 + 9}.$$

Theorem 9.7(a) gives $1/n^3 \to 0$ and $1/n^2 \to 0$. The finite scalar and sum laws therefore give $1 + 8/n^3 \to 1$ and $1 + 9/n^2 \to 1$. All the latter denominators are positive, and the limit denominator is $1 \neq 0$, so the finite quotient theorem applies to this fraction and gives

$$v_n = \frac{1 + 8/n^3}{1 + 9/n^2} \longrightarrow 1 > 0.$$

Also $n^2 \to +\infty$: for each $M > 0$, every $n > \sqrt{M}$ has $n^2 > M$. Thus Theorem 9.9 applies to $s_n = n^2$ and $t_n = v_n$, since its first limit is $+\infty$ and its second limit is the positive number 1. It gives the desired product limit.

Conclusion. In part (a), $\lim_{n\to\infty} \dfrac{n^4 + 8n}{n^2 + 9} = +\infty$.

(b) The expression $2^n/n^2 + (-1)^n$.

Exercise 9.14 with $a = 2 > 1$ and $p = 2 > 0$ gives

$$u_n = \frac{2^n}{n^2} \longrightarrow +\infty.$$

The second sequence is bounded: $-1 \leq (-1)^n \leq 1$ for every n. Exercise 9.11(c) therefore applies to the sum. In explicit threshold form, given $M > 0$, choose N so that $n > N$ implies $u_n > M + 1$. Then

$$\frac{2^n}{n^2} + (-1)^n \geq u_n - 1 > M.$$

This verifies the limit without requiring the oscillating sequence $(-1)^n$ itself to have a limit.

Conclusion. In part (b), $\lim_{n\to\infty} \left[\frac{2^n}{n^2} + (-1)^n\right] = +\infty$.

(c) The expression $3^n/n^3 - 3^n/n!$.

Exercise 9.14 with $a = 3$, $p = 3$ and Exercise 9.15 with $a = 3$ give, respectively,

$$u_n = \frac{3^n}{n^3} \longrightarrow +\infty, \qquad v_n = \frac{3^n}{n!} \longrightarrow 0.$$

Thus $-v_n \to 0$ by the finite scalar-multiple theorem. Exercise 9.11(b) applies to $u_n + (-v_n)$, since the second limit 0 is greater than $-\infty$. To verify the threshold directly, fix $M > 0$. Choose N_1 such that $n > N_1$ implies $u_n > M + 1$. Choose N_2 such that $n > N_2$ implies $|v_n| < 1$, using its convergence to 0. For $n > \max\{N_1, N_2\}$ we have $v_n < 1$ and hence

$$u_n - v_n > u_n - 1 > M.$$

This is a growing term minus a term tending to 0; the proof does not subtract two infinite limits.

Conclusion. In part (c), $\lim_{n\to\infty} \left[\frac{3^n}{n^3} - \frac{3^n}{n!}\right] = +\infty$.

9.17 Exercise 9.17: A definition-only proof for the sequence of squares

What we must show. Use only Definition 9.8 to prove $n^2 \to +\infty$.

Plan. Given the positive height M, solve $n^2 > M$ by requiring $n > \sqrt{M}$. Both sides are positive, so squaring this comparison preserves its direction.

Formal proof with the quantifiers explicit.

Let $M > 0$ be arbitrary, and set $N = \sqrt{M}$. If $n \in \mathbb{N}$ and $n > N$, then $n > \sqrt{M} > 0$. Multiplying positive inequalities, or using the strictly increasing square function on the positive numbers, gives

$$n^2 > (\sqrt{M})^2 = M.$$

More explicitly, $n > \sqrt{M}$ gives $n^2 > n\sqrt{M} > (\sqrt{M})^2$ after multiplication first by $n > 0$ and then by $\sqrt{M} > 0$. We have therefore proved that for every $M > 0$ there is $N = \sqrt{M}$ such that every positive integer $n > N$ satisfies $n^2 > M$. This is exactly Definition 9.8.

Conclusion. $\lim_{n\to\infty} n^2 = +\infty$, with the explicit threshold $N = \sqrt{M}$ for each $M > 0$.

9.18 Exercise 9.18: Finite geometric sums and their limits

What we must show. Prove the finite-sum identity in (a), then determine the limits in (b), (c), and (d), including the base $a = 1$ where the identity's quotient is not defined.

Plan. Subtract a times the finite sum from the original sum. Every interior power cancels. After the identity is proved, use the established geometric-sequence limit for $|a| < 1$, and a direct lower comparison for $a \geq 1$.

(a) Prove the identity for every allowed base and every index.

Fix a real $a \neq 1$ and a positive integer n, and define

$$S_n = 1 + a + a^2 + \cdots + a^n = \sum_{k=0}^{n} a^k.$$

Here a^0 denotes the constant empty product 1, including when $a = 0$. Multiplication and reindexing a finite sum give

$$aS_n = \sum_{k=0}^{n} a^{k+1} = \sum_{j=1}^{n+1} a^j = a + a^2 + \cdots + a^n + a^{n+1}.$$

The reindexing sets $j = k + 1$, so the lower bound $k = 0$ becomes $j = 1$ and the upper bound $k = n$ becomes $j = n + 1$. Subtracting the two finite sums cancels exactly the terms $a, a^2, \ldots, a^n$:

$$\begin{aligned} S_n - aS_n &= (1 + a + \cdots + a^n) - (a + \cdots + a^n + a^{n+1}) \\ &= 1 - a^{n+1}. \end{aligned}$$

The left side factors as $(1 - a)S_n$. Since $a \neq 1$, $1 - a \neq 0$, and division by this nonzero number yields

$$S_n = \frac{1 - a^{n+1}}{1 - a}.$$

The base and index were arbitrary, so this proves the identity for every real $a \neq 1$ and every positive integer n. The cancellation also works for $n = 0$, if that index is included, since then the identity is $1 = (1 - a)/(1 - a)$.

(b) Pass to the limit for $|a| < 1$.

Now fix a with $|a| < 1$. Then $a \neq 1$, so (a) applies. Exercise 9.13 gives $a^n \to 0$. Since $a^{n+1} = a\,a^n$, multiplication by the fixed real number a gives $a^{n+1} \to a \cdot 0 = 0$ by the finite scalar-multiple theorem. The denominator $1 - a$ is a fixed nonzero real number. Consequently the finite sum and quotient laws applied to the identity give

$$\lim_{n\to\infty} S_n = \frac{1 - 0}{1 - a} = \frac{1}{1 - a}.$$

One can read the same conclusion directly as an error estimate, avoiding any ambiguity about division:

$$\begin{aligned} \left| S_n - \frac{1}{1 - a} \right| &= \left| \frac{1 - a^{n+1} - 1}{1 - a} \right| \\ &= \frac{|a|^{n+1}}{|1 - a|} \longrightarrow 0. \end{aligned}$$

The denominator in this expression is positive and constant; the numerator tends to 0. This includes $a = 0$, when every sum is exactly 1.

Conclusion. For part (b), if $|a| < 1$, then $\lim_{n\to\infty}(1 + a + \cdots + a^n) = 1/(1 - a)$.

(c) Substitute $a = 1/3$ and simplify the fraction.

The displayed sequence of sums is

$$1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \cdots + \frac{1}{3^n} = \sum_{k=0}^{n} \left(\frac{1}{3}\right)^k.$$

Since $0 < 1/3 < 1$, part (b) applies. Its limit is

$$\frac{1}{1 - 1/3} = \frac{1}{(3 - 1)/3} = \frac{1}{2/3} = \frac{3}{2}.$$

Conclusion. The limit in part (c) is $3/2$.

(d) Handle $a = 1$ and $a > 1$ explicitly.

If $a = 1$, the sum contains $n + 1$ terms, each equal to 1, so $S_n = n + 1$. For an arbitrary $M > 0$, $n > M$ implies $S_n = n + 1 > M$. Therefore $S_n \to +\infty$ in this case. The quotient in (a) was not used at $a = 1$, where its denominator would be zero.

If $a > 1$, every term a^k for $k = 0, 1, \ldots, n$ is at least 1: the exponent 0 gives 1, and a positive exponent gives a product of factors greater than 1. Adding the $n + 1$ inequalities gives

$$S_n = \sum_{k=0}^{n} a^k \geq \sum_{k=0}^{n} 1 = n + 1.$$

Since $n + 1 \to +\infty$, Exercise 9.9(a) proves $S_n \to +\infty$. Equivalently, the same threshold $N = M$ works directly: $n > N$ implies $S_n \geq n + 1 > M$.

Conclusion. For part (d), every base $a \geq 1$ gives $\lim_{n\to\infty}(1 + a + \cdots + a^n) = +\infty$.

10 Monotone sequences and Cauchy sequences

An increasing sequence satisfies $s_n \le s_{n+1}$; a decreasing sequence satisfies $s_n \ge s_{n+1}$. Both definitions permit equality. A sequence is bounded if all its terms lie between two fixed real numbers.

The Cauchy condition says that, for every $\varepsilon > 0$, there is one threshold N such that

$$m > N,\ n > N \quad \Longrightarrow \quad |s_m - s_n| < \varepsilon.$$

It controls every pair of late terms. Merely making successive differences small is not enough, as Exercise 10.6 explains. The solutions first justify convergence and then identify a limit.

Conventions for monotonicity and Cauchy estimates

In Definition 10.1, "increasing" means $s_n \le s_{n+1}$ and "decreasing" means $s_n \ge s_{n+1}$. Equality is permitted in both definitions. A constant sequence therefore has both properties. Repeated application of an increasing inequality gives $s_n \le s_m$ whenever $n < m$; for a decreasing sequence it gives $s_n \ge s_m$.

Definition 10.8 says that (s_n) is Cauchy if

> for every $\varepsilon > 0$ there is an N such that every pair of integers $m, n > N$ satisfies $|s_m - s_n| < \varepsilon$.

Both indices are arbitrary after the same threshold. An estimate for consecutive terms alone is not yet the estimate for an arbitrary pair.

10.1 Exercise 10.1: Classifying six sequences

What we must show. Decide whether each sequence is increasing, decreasing, and bounded, with the weak monotonicity conventions just stated.

Part (a): the sequence $s_n = 1/n$.

For every $n \in \mathbb{N}$, the denominators n and $n+1$ are positive, and

$$s_{n+1} - s_n = \frac{1}{n+1} - \frac{1}{n} = \frac{n-(n+1)}{n(n+1)} = -\frac{1}{n(n+1)} < 0.$$

Thus $s_{n+1} < s_n$ for every n, so the sequence is decreasing. It is not increasing: for example, $s_2 = 1/2 < 1 = s_1$. Because $n \ge 1$, we also have $0 < 1/n \le 1$, so 0 and 1 are a lower and an upper bound.

Conclusion. The sequence $1/n$ is **decreasing and bounded**, and is not increasing.

Part (b), Step 1: test monotonicity with exact terms.

Here $s_n = (-1)^n/n^2$. Its first three terms are

$$s_1 = -1, \qquad s_2 = \frac{1}{4}, \qquad s_3 = -\frac{1}{9}.$$

The inequality $s_1 < s_2$ rules out a decreasing sequence, which would require $s_1 \ge s_2$. The inequality $s_2 > s_3$ rules out an increasing sequence, which would require $s_2 \le s_3$. Hence it is neither.

Part (b), Step 2: give bounds valid at every index.

If n is odd, then $s_n = -1/n^2$, so $-1 \le s_n < 0$. If n is even, then $n \ge 2$ and $s_n = 1/n^2$, so $0 < s_n \le 1/4$. Both cases give

$$-1 \le s_n \le \frac{1}{4} \qquad (n \in \mathbb{N}).$$

These bounds are attained at $n = 1$ and $n = 2$, respectively.

Conclusion. The sequence $(-1)^n/n^2$ is **bounded but neither increasing nor decreasing**.

Part (c), Step 1: check the growth of $s_n = n^5$.

For every $n \ge 1$, the binomial expansion gives

$$\begin{aligned} s_{n+1} - s_n &= (n+1)^5 - n^5, \\ &= n^5 + 5n^4 + 10n^3 + 10n^2 + 5n + 1 - n^5, \\ &= 5n^4 + 10n^3 + 10n^2 + 5n + 1 > 0. \end{aligned}$$

All five terms in the final sum are positive. Thus $s_{n+1} > s_n$ for every index, giving an increasing sequence and ruling out a decreasing one.

Part (c), Step 2: prove unboundedness above.

Since $n \ge 1$, we have $n^4 \ge 1$, so $n^5 \ge n \ge 1$. The number 1 is a lower bound. To show there is no upper bound, let $U \in \mathbb{R}$ be any proposed upper bound. By the Archimedean property, choose $n \in \mathbb{N}$ with $n > \max\{1, U\}$. Then

$$s_n = n^5 \ge n > U.$$

Thus U fails to bound the sequence above. This works for every real candidate U.

Conclusion. The sequence n^5 is **increasing and unbounded above**. It is not decreasing or bounded, although it is bounded below by 1.

Part (d), Step 1: give a rise and a fall for the sine sequence.

Let $s_n = \sin(n\pi/7)$. Put $c = \sin(\pi/7)$. Since $0 < \pi/7 < \pi$, the elementary sign property of sine gives $c > 0$. The sine values at three indices are

$$s_1 = c > 0, \qquad s_7 = \sin \pi = 0, \qquad s_{15} = \sin(2\pi + \pi/7) = c > 0.$$

An increasing sequence would satisfy $s_1 \le s_7$, contradicting $s_1 > 0 = s_7$. A decreasing sequence would satisfy $s_7 \ge s_{15}$, contradicting $s_7 = 0 < s_{15}$. Comparisons at nonadjacent indices are legitimate because monotonicity implies the corresponding comparison at any pair of ordered indices.

Part (d), Step 2: give global bounds.

The elementary trigonometric inequality $-1 \le \sin x \le 1$ is valid for every real x. Applying it at $x = n\pi/7$ gives $-1 \le s_n \le 1$ for every n.

Conclusion. The sequence $\sin(n\pi/7)$ is **bounded but neither increasing nor decreasing**.

Part (e), Step 1: rule out both monotonicity directions.

For $s_n = (-2)^n$, the first three terms are

$$s_1 = -2, \qquad s_2 = 4, \qquad s_3 = -8.$$

The rise from -2 to 4 rules out a decreasing sequence. The fall from 4 to -8 rules out an increasing sequence.

Part (e), Step 2: record an elementary power estimate.

We will use

$$2^j \geq j+1 \qquad (j \in \mathbb{N}).$$

For $j = 1$, this is $2 \geq 2$. If it holds for $j \geq 1$, then

$$2^{j+1} = 2 \cdot 2^j \geq 2(j+1) = 2j+2 \geq j+2,$$

since $(2j+2) - (j+2) = j \geq 0$. Induction proves the estimate.

Part (e), Step 3: show that both signs have arbitrarily large magnitude.

Let $R > 0$ be arbitrary. Choose $k \in \mathbb{N}$ with $k > R$ by the Archimedean property. At an even index,

$$s_{2k} = 2^{2k} \geq 2k+1 > R.$$

At the following odd index,

$$s_{2k+1} = -2^{2k+1} \leq -(2k+2) < -R.$$

Thus there are terms above R and below $-R$ for every positive R. In particular, no real number is an upper bound and no real number is a lower bound.

Conclusion. The sequence $(-2)^n$ is **neither increasing nor decreasing, and is unbounded both above and below**.

Part (f), Step 1: compare consecutive positive terms.

Here $s_n = n/3^n > 0$. Since we are dividing positive quantities, we can compute the consecutive-term ratio:

$$\begin{aligned} \frac{s_{n+1}}{s_n} &= \frac{n+1}{3^{n+1}} \frac{3^n}{n}, \\ &= \frac{n+1}{3n}. \end{aligned}$$

For $n \geq 1$, the denominator is positive and $3n-(n+1) = 2n-1 \geq 1 > 0$. Hence $0 < (n+1)/(3n) < 1$. Multiplying this strict upper inequality by $s_n > 0$ gives $s_{n+1} < s_n$.

Part (f), Step 2: bound the whole sequence.

Every term is positive. Repeated application of the decreasing inequality gives $s_n \leq s_1 = 1/3$. Therefore

$$0 < s_n \leq \frac{1}{3} \qquad (n \in \mathbb{N}).$$

The strict decrease also rules out an increasing sequence.

Conclusion. The sequence $n/3^n$ is **decreasing and bounded**, and is not increasing.

Summary of all six classifications.

Sequence	Increasing?	Decreasing?	Bounded?
$1/n$	No	Yes	Yes
$(-1)^n/n^2$	No	No	Yes
n^5	Yes	No	No
$\sin(n\pi/7)$	No	No	Yes
$(-2)^n$	No	No	No
$n/3^n$	No	Yes	Yes

10.2 Exercise 10.2: Convergence of a bounded decreasing sequence

What we must show. Prove the decreasing case of Theorem 10.2: every bounded decreasing real sequence converges.

Step 1: define the candidate limit and justify its existence.

Let (s_n) be bounded and decreasing, and let

$$S = \{s_n : n \in \mathbb{N}\}.$$

The set is nonempty because $s_1 \in S$, and it is bounded because the sequence is bounded. In particular, it is bounded below. The lower-bound form of completeness gives a real number

$$L = \inf S.$$

Being a lower bound means $L \leq s_n$ for every n.

Step 2: obtain one term close to the infimum.

Let $\varepsilon > 0$. The number $L + \varepsilon$ is strictly larger than the greatest lower bound L, so it cannot itself be a lower bound of S. Therefore there is an element of S below $L + \varepsilon$. By the definition of S, that element is s_N for some index $N \in \mathbb{N}$. Thus

$$L \leq s_N < L + \varepsilon.$$

Step 3: use decrease to control every subsequent term.

Take any integer $n > N$. Since the sequence is decreasing, $s_n \leq s_N$. Since L is a lower bound, $L \leq s_n$. Combining these facts with Step 2 gives

$$L \leq s_n \leq s_N < L + \varepsilon.$$

Subtract L throughout to get $0 \leq s_n - L < \varepsilon$. The difference $s_n - L$ is nonnegative, so

$$|s_n - L| = s_n - L < \varepsilon \qquad \text{for every integer } n > N.$$

Conclusion. For every $\varepsilon > 0$ we obtained the threshold required by Definition 7.1. Hence $\boxed{s_n \longrightarrow \inf\{s_n : n \in \mathbb{N}\}}$. The proof uses the greatest-lower-bound property, without assuming the infimum is one of the terms.

10.3 Exercise 10.3: A finite bound for decimal approximations

What we must show. Let K be a nonnegative integer and let each digit d_j belong to $\{0, 1, \ldots, 9\}$. For

$$s_n = K + \sum_{j=1}^{n} \frac{d_j}{10^j},$$

prove $s_n < K + 1$ for every $n \in \mathbb{N}$.

Step 1: bound each digit contribution.

For every $j \geq 1$, $0 \leq d_j \leq 9$ and $10^j > 0$. Division by the positive denominator gives

$$0 \leq \frac{d_j}{10^j} \leq \frac{9}{10^j}.$$

Add these finitely many inequalities and then add K:

$$K \leq s_n \leq K + \sum_{j=1}^{n} \frac{9}{10^j}.$$

Step 2: evaluate the finite upper-bound sum by cancellation.

For an individual index j,

$$\frac{9}{10^j} = \frac{10-1}{10^j} = \frac{1}{10^{j-1}} - \frac{1}{10^j}.$$

Consequently

$$\begin{aligned} \sum_{j=1}^{n} \frac{9}{10^j} &= \left(1 - \frac{1}{10}\right) + \left(\frac{1}{10} - \frac{1}{10^2}\right) + \cdots + \left(\frac{1}{10^{n-1}} - \frac{1}{10^n}\right), \\ &= 1 - \frac{1}{10^n}. \end{aligned}$$

Every intermediate negative term cancels the identical positive term in the next parentheses. This is a finite algebraic identity; no limit or infinite sum has been used to derive it.

Step 3: preserve the strict final inequality.

Substitute the identity from Step 2 into Step 1:

$$K \leq s_n \leq K + 1 - \frac{1}{10^n}.$$

Since $10^n > 0$, its reciprocal is strictly positive. Subtracting that positive number makes $K+1-1/10^n$ strictly less than $K + 1$. Thus

$$s_n \leq K + 1 - \frac{1}{10^n} < K + 1.$$

Conclusion. Every finite decimal approximation satisfies $\boxed{s_n < K + 1}$. Moreover $s_{n+1} - s_n = d_{n+1}/10^{n+1} \geq 0$, so the approximations are increasing and bounded. Their convergence follows from Theorem 10.2 *after* the finite bound has been established.

10.4 Exercise 10.4: Why the monotone and Cauchy theorems require real completeness

What we must show. Explain why Theorems 10.2 and 10.11 would be false if the permitted terms and limits were restricted to $\mathbb{Q}$.

Plan. We construct a sequence of rational decimal truncations to $\sqrt{2}$. It is increasing, bounded by rational numbers, and Cauchy, but its only real limit is the irrational number $\sqrt{2}$. The construction itself uses finite sets and rational comparisons, rather than assuming that $\mathbb{Q}$ has the completeness property being tested.

Step 1: construct each truncation using integers only.

For each $n \in \mathbb{N}$, consider the finite set

$$E_n = \{k \in \mathbb{Z} : 0 \le k \le 2 \cdot 10^n,\ k^2 < 2 \cdot 10^{2n}\}.$$

It is nonempty because $0 \in E_n$. A finite nonempty set of integers has a largest member, so define

$$k_n = \max E_n, \qquad q_n = \frac{k_n}{10^n}.$$

The denominator is a positive integer, and the numerator is an integer. Hence every q_n is rational. The operations and comparisons used to define E_n involve only integers. In particular, this construction does not presuppose a supremum in $\mathbb{Q}$ for an infinite set.

Step 2: prove rational lower and upper bounds.

The integer 10^n belongs to E_n, since it is between 0 and $2 \cdot 10^n$ and its square 10^{2n} is strictly less than $2 \cdot 10^{2n}$. Thus $k_n \ge 10^n$. The integer $2 \cdot 10^n$ does not belong to E_n, since its square is $4 \cdot 10^{2n} > 2 \cdot 10^{2n}$. Therefore $k_n < 2 \cdot 10^n$. Dividing by $10^n > 0$ gives

$$1 \le q_n < 2.$$

Both bounds are rational, so the sequence is bounded even when all bounding numbers are required to lie in $\mathbb{Q}$.

Step 3: prove that the sequence is increasing.

Since $k_n^2 < 2 \cdot 10^{2n}$, multiplication by 100 gives

$$(10k_n)^2 < 2 \cdot 10^{2(n+1)}.$$

Also $0 \le 10k_n \le 2 \cdot 10^{n+1}$. Thus $10k_n \in E_{n+1}$. The maximality of k_{n+1} implies $10k_n \le k_{n+1}$. Dividing by $10^{n+1} > 0$ yields

$$q_n = \frac{10k_n}{10^{n+1}} \le \frac{k_{n+1}}{10^{n+1}} = q_{n+1}.$$

Equality may occur at some indices, which is permitted by the definition of increasing.

Step 4: give a rational upper approximation to the same square.

The integer $k_n + 1$ is still at most $2 \cdot 10^n$, because $k_n < 2 \cdot 10^n$ and both numbers are integers. If $(k_n + 1)^2 < 2 \cdot 10^{2n}$, then $k_n + 1$ would belong to E_n, contradicting maximality. Therefore

$$k_n^2 < 2 \cdot 10^{2n} \le (k_n + 1)^2.$$

Divide by $10^{2n} > 0$. The result is

$$q_n^2 < 2 \leq \left(q_n + \frac{1}{10^n}\right)^2.$$

Every number in this display is rational, so the comparisons remain meaningful entirely within $\mathbb{Q}$.

Step 5: verify the Cauchy condition through rational estimates.

For integers $m > n$, Step 3 gives $q_n \leq q_m$. The positive numbers q_m and $q_n + 10^{-n}$ satisfy

$$q_m^2 < 2 \leq (q_n + 10^{-n})^2.$$

Hence $q_m < q_n + 10^{-n}$. Indeed, the contrary inequality $q_m \geq q_n + 10^{-n} \geq 0$ would imply the contrary comparison between their squares. We conclude

$$0 \leq q_m - q_n < 10^{-n} \qquad (m > n).$$

We may use $10^j \geq j$ for every $j \in \mathbb{N}$: the case $j = 1$ is $10 \geq 1$; if $10^j \geq j$, then $10^{j+1} \geq 10j \geq j + 1$ because $j \geq 1$. This proves the estimate by induction, and taking reciprocals gives $10^{-j} \leq 1/j$.

Now let $\varepsilon > 0$ and choose $N = \max\{1, 1/\varepsilon\}$. Take any two integer indices $m, n > N$. If $m = n$, then $|q_m - q_n| = 0 < \varepsilon$. If $m > n$, the preceding estimate gives

$$|q_m - q_n| = q_m - q_n < 10^{-n} \leq \frac{1}{n} < \varepsilon.$$

If $n > m$, interchange their roles and use the symmetry of absolute value to obtain $|q_m - q_n| < 10^{-m} \leq 1/m < \varepsilon$. Thus the Cauchy estimate holds for every pair of indices above N. In particular, it holds for every positive rational tolerance, as a Cauchy condition formulated within $\mathbb{Q}$ requires.

Step 6: identify the only possible real limit.

In the ambient real numbers, let $\alpha = \sqrt{2}$. It is positive, $\alpha^2 = 2$, and it is irrational by Section 2. Step 4 and comparison of squares of positive numbers give

$$q_n < \alpha \leq q_n + 10^{-n}.$$

The upper equality is impossible because $q_n + 10^{-n}$ is rational and α is irrational. Therefore

$$0 < \alpha - q_n < 10^{-n} \leq \frac{1}{n}.$$

For any real $\varepsilon > 0$, the choice $N = \max\{1, 1/\varepsilon\}$ makes the right side strictly smaller than ε whenever $n > N$. Hence Definition 7.1 gives $q_n \to \alpha$ as a sequence in $\mathbb{R}$.

Step 7: rule out every rational limit.

A limit in $\mathbb{Q}$ would have to be some $r \in \mathbb{Q}$. Restricting the definition to positive rational tolerances makes no difference: given any real tolerance $\varepsilon > 0$, the Archimedean property gives a positive rational $1/j < \varepsilon$, and the estimate for that smaller rational tolerance implies the estimate for ε. Thus a limit in $\mathbb{Q}$ would also be the same limit in $\mathbb{R}$.

Because r is rational and α is irrational, they are distinct. Put $d = |\alpha - r| > 0$. If q_n converged to both numbers, there would be thresholds N_1, N_2 such that, for $n > \max\{N_1, N_2\}$,

$$|q_n - \alpha| < \frac{d}{3}, \qquad |q_n - r| < \frac{d}{3}.$$

The triangle inequality would then give

$$d = |\alpha - r| \leq |\alpha - q_n| + |q_n - r| < \frac{2d}{3} < d,$$

a contradiction. Thus the rational sequence has no rational limit.

Conclusion. The same rational sequence is bounded and increasing, so it contradicts the version of Theorem 10.2 with limits restricted to $\mathbb{Q}$. It is also Cauchy, so it contradicts the corresponding version of Theorem 10.11. Both failures arise because the required limit $\sqrt{2}$ is missing from $\mathbb{Q}$, not because the sequence is unbounded or fails its Cauchy estimates.

10.5 Exercise 10.5: An unbounded decreasing sequence tends to negative infinity

What we must show. Prove Theorem 10.4(ii): if a real sequence (s_n) is decreasing and unbounded, then $s_n \to -\infty$.

Step 1: determine which direction must be unbounded.

Since the sequence is decreasing, every term satisfies $s_n \leq s_1$. Thus s_1 is an upper bound of its set of terms. If the sequence also had a real lower bound, it would be bounded above and below, contrary to the hypothesis that it is unbounded. Therefore the sequence is unbounded below.

Step 2: find a term below an arbitrary target.

Let $M < 0$ be any real target. Since the sequence is unbounded below, M cannot be a lower bound for all its terms. Hence some index $N \in \mathbb{N}$ satisfies

$$s_N < M.$$

Step 3: show that every later term remains below that target.

For every integer $n > N$, decrease gives $s_n \leq s_N$. Thus

$$s_n \leq s_N < M \qquad (n > N).$$

We have proved that for each $M < 0$ there exists a threshold N such that every later term is less than M.

Conclusion. This is the definition of divergence to negative infinity, so $\boxed{s_n \longrightarrow -\infty}$. The choice of s_N would in fact work for any real target M, not just negative ones.

10.6 Exercise 10.6: Small successive differences: one sufficient rate and one insufficient rate

What we must show. Prove (a) that $|s_{n+1} - s_n| < 2^{-n}$ for every natural n implies (s_n) is Cauchy and convergent; and (b) decide whether the same conclusion follows from the weaker estimate $|s_{n+1} - s_n| < 1/n$.

Part (a), Step 1: compare arbitrary terms by telescoping.

Fix integers $m > n$. Then

$$s_m - s_n = \sum_{k=n}^{m-1} (s_{k+1} - s_k).$$

To see the cancellation, expand the right side as $(s_{n+1} - s_n) + (s_{n+2} - s_{n+1}) + \cdots + (s_m - s_{m-1})$; every intermediate term occurs once with each sign. Applying the finite triangle inequality gives

$$|s_m - s_n| \leq \sum_{k=n}^{m-1} |s_{k+1} - s_k| < \sum_{k=n}^{m-1} 2^{-k}.$$

The final inequality is strict because the finite sum is nonempty and every one of its summands is strictly less than the corresponding 2^{-k}.

Part (a), Step 2: bound this finite geometric sum.

For an individual k, $2^{-k} = 2^{1-k} - 2^{-k}$. Substituting that identity into the sum makes the consecutive terms telescope:

$$\begin{aligned}\sum_{k=n}^{m-1} 2^{-k} &= (2^{1-n} - 2^{-n}) + (2^{-n} - 2^{-n-1}) + \cdots + (2^{2-m} - 2^{1-m}),\\ &= 2^{1-n} - 2^{1-m},\\ &< 2^{1-n}.\end{aligned}$$

The last inequality holds because $2^{1-m} > 0$. We have therefore shown

$$|s_m - s_n| < 2^{1-n} \qquad (m > n).$$

Part (a), Step 3: choose one threshold for all pairs.

Let $\varepsilon > 0$. By the Archimedean property choose $N \in \mathbb{N}$ such that

$$N > \max\left\{1, \frac{2}{\varepsilon}\right\}.$$

The elementary estimate $2^N \geq N + 1$ was proved in Exercise 10.1(e). It gives

$$2^{1-N} = \frac{2}{2^N} \leq \frac{2}{N+1} < \frac{2}{N} < \varepsilon.$$

Now let $m, n > N$ be arbitrary integer indices. If $m = n$, their terms have distance $0 < \varepsilon$. If $m > n$, Steps 1–2 give

$$|s_m - s_n| < 2^{1-n} \leq 2^{1-N} < \varepsilon.$$

If $n > m$, interchange the indices and use $|s_m - s_n| = |s_n - s_m|$ to get the same conclusion with 2^{1-m} in place of 2^{1-n}.

Conclusion. For every $\varepsilon > 0$ all pairs of terms after the chosen threshold have distance less than ε. Thus (s_n) is Cauchy by Definition 10.8, and Theorem 10.11 for real sequences implies that it converges to a real number.

Part (b), Step 1: construct a sequence satisfying the strict hypothesis.

Define the harmonic partial sums

$$H_n = \sum_{k=1}^{n} \frac{1}{k} = 1 + \frac{1}{2} + \cdots + \frac{1}{n}.$$

The consecutive difference is exactly

$$H_{n+1} - H_n = \frac{1}{n+1}.$$

Both sides are positive, and $n + 1 > n > 0$, so

$$|H_{n+1} - H_n| = \frac{1}{n+1} < \frac{1}{n} \qquad (n \in \mathbb{N}).$$

The required inequality is strict at every index, including $n = 1$.

Part (b), Step 2: show directly that the sequence is not Cauchy.

Fix the positive tolerance $\varepsilon_0 = 1/2$. Given any proposed threshold N, choose an integer $n > \max\{N, 1\}$ and put $m = 2n$. Both indices are greater than N. The difference of the terms is

$$H_{2n} - H_n = \sum_{k=n+1}^{2n} \frac{1}{k}.$$

There are exactly n summands. For each index in this sum, $k \leq 2n$, so $1/k \geq 1/(2n)$. Adding the n inequalities gives

$$|H_{2n} - H_n| = H_{2n} - H_n \geq n\frac{1}{2n} = \frac{1}{2} = \varepsilon_0.$$

Thus, no matter how N is chosen, some pair of indices $m, n > N$ violates the required strict Cauchy estimate.

Conclusion. The answer to part (b) is **no**. The sequence (H_n) satisfies every strict bound $|H_{n+1} - H_n| < 1/n$, but it is not Cauchy. Since every convergent sequence is Cauchy by Lemma 10.9, it does not converge.

10.7 Exercise 10.7: Approaching an unattained supremum by points of the set

What we must show. Let $S \subseteq \mathbb{R}$ be bounded and nonempty, and suppose $\sup S \notin S$. Construct a sequence (s_n) with $s_n \in S$ for every n and $s_n \to \sup S$.

Step 1: use the least-upper-bound property at each precision.

Completeness gives the real number $M = \sup S$. For each $n \in \mathbb{N}$, we have $1/n > 0$, and consequently $M - 1/n < M$. The smaller number $M - 1/n$ cannot be an upper bound of S, because M is its least upper bound. Therefore there is a point $s_n \in S$ satisfying $M - 1/n < s_n$. Since M is an upper bound, also $s_n \leq M$. Equality is ruled out by $M \notin S$. Thus, choosing such a point for each n, we obtain

$$M - \frac{1}{n} < s_n < M \qquad (n \in \mathbb{N}).$$

Step 2: translate the order estimate into an error estimate.

Subtracting s_n and rearranging the left inequality gives $M - s_n < 1/n$. The right inequality gives $M - s_n > 0$, so

$$|s_n - M| = M - s_n < \frac{1}{n}.$$

This bound holds at every index; the actual choices of the points s_n need not produce a monotone sequence.

Step 3: verify convergence with an explicit threshold.

Let $\varepsilon > 0$ and choose

$$N = \max\left\{1, \frac{1}{\varepsilon}\right\}.$$

For every integer $n > N$, $n > 1/\varepsilon$ gives $1/n < \varepsilon$, and therefore

$$|s_n - M| < \frac{1}{n} < \varepsilon.$$

Conclusion. The constructed sequence consists entirely of points of S and satisfies $\boxed{s_n \longrightarrow \sup S}$. The supremum itself is excluded from the set, so each approximation is strictly below its target.

10.8 Exercise 10.8: Averages of an increasing positive sequence

What we must show. Let (s_n) be increasing with $s_n > 0$ for every n, and let

$$\sigma_n = \frac{1}{n}\sum_{k=1}^{n} s_k.$$

Prove that the average sequence (σ_n) is increasing in the sense of Definition 10.1.

Step 1: compare each average with the last term it averages.

Fix $n \in \mathbb{N}$. Since (s_n) is increasing, every k with $1 \le k \le n$ satisfies $s_k \le s_n$. Add these n inequalities:

$$s_1 + s_2 + \cdots + s_n \le \underbrace{s_n + s_n + \cdots + s_n}_{n \text{ terms}} = ns_n.$$

Divide by the positive integer n to obtain $\sigma_n \le s_n$. Since $s_n \le s_{n+1}$, transitivity gives

$$\sigma_n \le s_n \le s_{n+1}.$$

In particular, $s_{n+1} - \sigma_n \ge 0$.

Step 2: express the next average in terms of the previous one.

The definition of σ_n gives $s_1 + \cdots + s_n = n\sigma_n$. Therefore

$$\begin{aligned}\sigma_{n+1} &= \frac{s_1 + \cdots + s_n + s_{n+1}}{n+1},\\ &= \frac{n\sigma_n + s_{n+1}}{n+1}.\end{aligned}$$

Step 3: calculate the exact difference between consecutive averages.

Put the terms over the same positive denominator $n+1$:

$$\begin{aligned}\sigma_{n+1}-\sigma_n &= \frac{n\sigma_n+s_{n+1}}{n+1}-\frac{(n+1)\sigma_n}{n+1},\\ &= \frac{n\sigma_n+s_{n+1}-(n+1)\sigma_n}{n+1},\\ &= \frac{n\sigma_n+s_{n+1}-n\sigma_n-\sigma_n}{n+1},\\ &= \frac{s_{n+1}-\sigma_n}{n+1}.\end{aligned}$$

The numerator is nonnegative by Step 1, and the denominator $n+1$ is positive. Hence $\sigma_{n+1}-\sigma_n \geq 0$, or $\sigma_n \leq \sigma_{n+1}$.

Conclusion. This comparison holds for every $n \in \mathbb{N}$, proving $\boxed{(\sigma_n) \text{ is increasing}}$. Equality can occur: for a constant positive sequence, all the averages are equal. Thus the weak meaning of "increasing" is the one required by the hypotheses and conclusion.

10.9 Exercise 10.9: A decreasing recurrence with a squared term

What we must show. For

$$s_1 = 1, \qquad s_{n+1} = \frac{n}{n+1}s_n^2 \quad (n \in \mathbb{N}),$$

compute s_2, s_3, s_4, prove that a finite limit exists, and prove that its value is 0.

Part (a): substitute the indices one at a time.

For $n = 1$, the recurrence gives

$$s_2 = \frac{1}{1+1}s_1^2 = \frac{1}{2}\cdot 1^2 = \frac{1}{2}.$$

For $n = 2$,

$$s_3 = \frac{2}{2+1}s_2^2 = \frac{2}{3}\left(\frac{1}{2}\right)^2 = \frac{2}{3}\cdot\frac{1}{4} = \frac{1}{6}.$$

For $n = 3$,

$$s_4 = \frac{3}{3+1}s_3^2 = \frac{3}{4}\left(\frac{1}{6}\right)^2 = \frac{3}{4}\cdot\frac{1}{36} = \frac{3}{144} = \frac{1}{48}.$$

Conclusion. $\boxed{s_2 = \frac{1}{2}, \quad s_3 = \frac{1}{6}, \quad s_4 = \frac{1}{48}}$.

Part (b), Step 1: prove positivity and an upper bound by induction.

We claim $0 < s_n \leq 1$ for every $n \in \mathbb{N}$.

The initial case is $s_1 = 1$, so $0 < s_1 \leq 1$ holds. For the induction step, assume $0 < s_n \leq 1$. Since $n \geq 1$, the coefficient satisfies

$$0 < \frac{n}{n+1} < 1.$$

Also $0 < s_n^2 \leq 1$. Multiplying these positive quantities gives

$$0 < s_{n+1} = \frac{n}{n+1}s_n^2 < 1.$$

This proves the claim by induction.

Part (b), Step 2: compare consecutive terms.

The inequality $s_n \le 1$, multiplied by the positive number s_n, gives $s_n^2 \le s_n$. The coefficient $n/(n+1)$ is strictly below 1, and $s_n^2 > 0$, so

$$s_{n+1} = \frac{n}{n+1} s_n^2 < s_n^2 \le s_n.$$

Hence the sequence is strictly decreasing. It is bounded below by 0 and above by 1.

Part (b), Step 3: apply monotone convergence before manipulating a limit.

Theorem 10.2 for bounded monotone sequences now applies. There is a real number L with $s_n \to L$. The inequalities $s_n > 0$ imply $L \ge 0$ by Exercise 8.9(a). Since the sequence is decreasing and $s_2 = 1/2$,

$$0 < s_n \le \frac{1}{2} \qquad (n \ge 2).$$

Exercise 8.9 also passes this eventual upper bound to the limit. Therefore

$$0 \le L \le \frac{1}{2}.$$

Conclusion. A finite limit exists, and it belongs to $[0, 1/2]$. Its existence has been proved independently of any equation for its value.

Part (c), Step 1: justify the shifted limit and the coefficient limit.

If $s_n \to L$, then $s_{n+1} \to L$ as well. Indeed, for any $\varepsilon > 0$ choose a threshold N with $m > N \Rightarrow |s_m - L| < \varepsilon$. For $n > N$, we have $n + 1 > N$, so the same estimate holds with $m = n + 1$.

Next put $c_n = n/(n+1)$. Its error from 1 is

$$|c_n - 1| = \left| \frac{n - (n+1)}{n+1} \right| = \frac{1}{n+1}.$$

For any $\varepsilon > 0$, if $n > 1/\varepsilon$, then $n + 1 > 1/\varepsilon$, so $1/(n+1) < \varepsilon$. Thus $c_n \to 1$.

Part (c), Step 2: apply the product limit law to the established limits.

Because $s_n \to L$ and $c_n \to 1$, the product laws from Section 9 give

$$\lim_{n\to\infty} c_n s_n^2 = 1 \cdot L \cdot L = L^2.$$

The recurrence says that this is the same sequence as s_{n+1}, whose limit is L. Uniqueness of a finite limit gives

$$L = L^2.$$

Rearranging and factoring gives

$$L^2 - L = 0, \qquad L(L-1) = 0.$$

The zero-product property therefore gives $L = 0$ or $L = 1$.

Part (c), Step 3: use the tail bound to remove the extraneous candidate.

Part (b) proved $L \le 1/2$, so $L = 1$ is impossible. The only remaining candidate is $L = 0$.

Conclusion. $\boxed{\lim_{n\to\infty} s_n = 0}$.

An additional direct error bound.

The same recurrence supplies a quantitative check. Since $s_n^2 \le s_n$,

$$s_{n+1} \le \frac{n}{n+1} s_n.$$

Starting with $s_1 = 1 = 1/1$, induction gives $s_n \le 1/n$: if $s_n \le 1/n$, then

$$s_{n+1} \le \frac{n}{n+1} \cdot \frac{1}{n} = \frac{1}{n+1}.$$

Consequently

$$|s_n - 0| = s_n \le \frac{1}{n} < \varepsilon \qquad \left(n > \frac{1}{\varepsilon}\right).$$

This also verifies the zero limit directly from Definition 7.1.

10.10 Exercise 10.10: A linear recurrence approaching one half

What we must show. For

$$s_1 = 1, \qquad s_{n+1} = \frac{s_n + 1}{3} \quad (n \in \mathbb{N}),$$

compute s_2, s_3, s_4, prove $s_n > 1/2$ by induction, prove that the sequence is decreasing, and identify its limit.

Part (a): calculate the first three new terms.

Using the recurrence successively gives

$$s_2 = \frac{1+1}{3} = \frac{2}{3},$$

$$s_3 = \frac{\frac{2}{3}+1}{3} = \frac{\frac{2}{3}+\frac{3}{3}}{3} = \frac{5/3}{3} = \frac{5}{9},$$

and

$$s_4 = \frac{\frac{5}{9}+1}{3} = \frac{\frac{5}{9}+\frac{9}{9}}{3} = \frac{14/9}{3} = \frac{14}{27}.$$

Conclusion. $\boxed{s_2 = \frac{2}{3}, \quad s_3 = \frac{5}{9}, \quad s_4 = \frac{14}{27}}$.

Part (b), Step 1: verify the initial case of the proposed lower bound.

The proposition for induction is P_n: $s_n > 1/2$. For $n = 1$, it says $1 > 1/2$, which is true.

Part (b), Step 2: prove the induction implication.

Assume $s_n > 1/2$. Adding 1 to both sides gives $s_n + 1 > 3/2$. Dividing by the positive number 3 preserves the strict inequality:

$$s_{n+1} = \frac{s_n + 1}{3} > \frac{3/2}{3} = \frac{1}{2}.$$

Thus P_n implies P_{n+1}. Induction proves the bound for every positive integer n.

Conclusion. $\boxed{s_n > \frac{1}{2} \text{ for every } n \in \mathbb{N}}$.

Part (c): subtract consecutive terms and determine the sign.

Use a common denominator:

$$s_{n+1} - s_n = \frac{s_n + 1}{3} - s_n = \frac{s_n + 1 - 3s_n}{3} = \frac{1 - 2s_n}{3}.$$

Part (b) gives $2s_n > 1$, so the numerator is negative. The denominator 3 is positive. Hence $s_{n+1} - s_n < 0$, which means $s_{n+1} < s_n$ for every n.

Conclusion. The sequence is strictly decreasing.

Part (d), Step 1: establish convergence from bounds and monotonicity.

The terms satisfy $1/2 < s_n \leq s_1 = 1$, and part (c) proves that the sequence is decreasing. Theorem 10.2 therefore gives a finite limit L. Exercise 8.9 implies

$$\frac{1}{2} \leq L \leq 1.$$

Part (d), Step 2: solve the limit equation after convergence is known.

The shifted sequence s_{n+1} has the same limit L, by the threshold argument given in Exercise 10.9(c). The addition and scalar-multiplication laws from Section 9 give

$$\lim_{n\to\infty} \frac{s_n + 1}{3} = \frac{L + 1}{3}.$$

The recurrence identifies these two sequences, so their limits agree:

$$L = \frac{L + 1}{3}.$$

Multiplying by 3 and collecting terms gives

$$3L = L + 1, \qquad 2L = 1, \qquad L = \frac{1}{2}.$$

Conclusion. $\boxed{\lim_{n\to\infty} s_n = \frac{1}{2}}$. Every term is strictly greater than $1/2$, but the limit can equal that lower bound.

10.11 Exercise 10.11: An infinite-product recurrence

What we must show. For

$$t_1 = 1, \qquad t_{n+1} = \left(1 - \frac{1}{4n^2}\right) t_n \quad (n \in \mathbb{N}),$$

(a) prove that a finite limit exists and (b) investigate its value. Part (b) asks what the limit appears to be. We will distinguish a numerical guess from a proof; an optional elementary-calculus argument will then verify the exact value.

Part (a), Step 1: bound the coefficient and prove positivity.

For every $n \geq 1$, we have $4n^2 \geq 4$, so

$$0 < \frac{1}{4n^2} \leq \frac{1}{4}.$$

Subtracting from 1 gives

$$\frac{3}{4} \leq 1 - \frac{1}{4n^2} < 1.$$

In particular, every recurrence coefficient is strictly positive. The initial value is $t_1 = 1 > 0$. If $t_n > 0$, the recurrence is a product of two positive numbers and gives $t_{n+1} > 0$. Induction shows $t_n > 0$ for all n.

Part (a), Step 2: prove monotonicity and apply the theorem.

Since $1 - 1/(4n^2) < 1$ and $t_n > 0$,

$$t_{n+1} = \left(1 - \frac{1}{4n^2}\right) t_n < t_n.$$

Thus (t_n) is strictly decreasing, bounded below by 0 and above by its first term 1. Theorem 10.2 gives a finite real limit L. The bound on the terms implies $0 \leq L \leq 1$ by Exercise 8.9.

Conclusion. The limit of (t_n) exists as a real number. This conclusion uses only positivity, monotonicity, and boundedness.

Part (b), Step 1: compute initial values and a numerical table.

The first terms are

$$t_2 = \left(1 - \frac{1}{4}\right) 1 = \frac{3}{4},$$

$$t_3 = \left(1 - \frac{1}{16}\right) \frac{3}{4} = \frac{15}{16} \cdot \frac{3}{4} = \frac{45}{64},$$

and

$$t_4 = \left(1 - \frac{1}{36}\right) \frac{45}{64} = \frac{35}{36} \cdot \frac{45}{64} = \frac{1575}{2304} = \frac{175}{256}.$$

Continuing the recurrence produces the following rounded values:

n	t_n
1	1.000000000000
2	0.750000000000
3	0.703125000000
4	0.683593750000
10	0.653587392299
100	0.638221323951
1000	0.636779026837

The numbers suggest a limit near 0.63662. The familiar constant $2/\pi$ has the decimal value $0.636619772368\ldots$, which suggests the guess

$$L = \frac{2}{\pi}.$$

The numerical agreement motivates this guess; finitely many numerical calculations by themselves do not prove the exact identification.

Part (b), Step 2: explain why a direct limit equation gives no value.

The coefficient $1 - 1/(4n^2)$ tends to 1. Taking limits in the recurrence, after convergence has been proved, only yields

$$L = 1 \cdot L.$$

Every real L satisfies this identity, so it does not determine the limit. Further information is needed to justify the guess $2/\pi$.

Optional verification of the guess: the elementary-calculus input.

The remaining argument uses integration by parts and the order property of definite integrals. Those are additional calculus tools; they were not needed for the existence proof in part (a). We derive the required product formula here rather than assume Wallis's product.

For an integer $m \geq 0$, define

$$I_m = \int_0^{\pi/2} \sin^m x \, dx,$$

where $\sin^0 x$ means the constant function 1. The integrands are continuous and are positive in the interior when $m \geq 1$, so all I_m are positive. Direct integration gives

$$I_0 = \frac{\pi}{2}, \qquad I_1 = 1.$$

Optional verification, Step 1: derive the integral recurrence.

For $m \geq 2$, write $I_m = \int_0^{\pi/2} \sin^{m-1} x \sin x \, dx$. In integration by parts take

$$u = \sin^{m-1} x, \quad dv = \sin x \, dx, \quad du = (m-1)\sin^{m-2} x \cos x \, dx, \quad v = -\cos x.$$

The boundary term $[-\sin^{m-1} x \cos x]_0^{\pi/2}$ is zero: $\cos(\pi/2) = 0$, and at 0 the factor $\sin^{m-1} 0$ is zero because $m - 1 \geq 1$. Consequently

$$\begin{aligned} I_m &= (m-1) \int_0^{\pi/2} \sin^{m-2} x \cos^2 x \, dx \\ &= (m-1) \int_0^{\pi/2} \sin^{m-2} x (1 - \sin^2 x) \, dx \\ &= (m-1)(I_{m-2} - I_m). \end{aligned}$$

Adding $(m-1)I_m$ to both sides and dividing by $m > 0$ gives

$$I_m = \frac{m-1}{m} I_{m-2}.$$

Optional verification, Step 2: express the even and odd integrals.

Repeated use of the recurrence, ending at I_0 or I_1, gives for every integer $N \geq 1$,

$$I_{2N} = \frac{\pi}{2} \prod_{k=1}^{N} \frac{2k-1}{2k}, \qquad I_{2N+1} = \prod_{k=1}^{N} \frac{2k}{2k+1}.$$

For example, the even recurrence successively contributes the factors $(2N-1)/(2N)$, $(2N-3)/(2N-2)$, down to $1/2$. The odd recurrence contributes $(2N)/(2N+1)$ down to $2/3$. These are exactly the displayed finite products.

Unrolling the recurrence for t_n gives

$$t_{N+1} = \prod_{k=1}^{N} \left(1 - \frac{1}{4k^2}\right) = \prod_{k=1}^{N} \frac{(2k-1)(2k+1)}{(2k)^2}.$$

The initial empty product for t_1 is 1, and each new recurrence step multiplies by the next factor, which justifies this finite-product identity. Dividing the formulas for the positive integrals gives

$$\frac{I_{2N}}{I_{2N+1}} = \frac{\pi}{2} \prod_{k=1}^{N} \frac{(2k-1)(2k+1)}{(2k)^2} = \frac{\pi}{2}\, t_{N+1}.$$

Optional verification, Step 3: bound the integral ratio.

On $[0, \pi/2]$, the quantity $\sin x$ lies in $[0, 1]$. Multiplication by a number in $[0, 1]$ cannot increase a nonnegative power, so

$$\sin^{2N+1} x \le \sin^{2N} x \le \sin^{2N-1} x.$$

Integrating preserves these inequalities:

$$I_{2N+1} \le I_{2N} \le I_{2N-1}.$$

Divide by $I_{2N+1} > 0$. The integral recurrence with $m = 2N + 1$ gives $I_{2N+1} = (2N)/(2N + 1)\, I_{2N-1}$, so

$$1 \le \frac{I_{2N}}{I_{2N+1}} \le \frac{I_{2N-1}}{I_{2N+1}} = \frac{2N+1}{2N} = 1 + \frac{1}{2N}.$$

Substituting $I_{2N}/I_{2N+1} = (\pi/2)t_{N+1}$ and multiplying by $2/\pi > 0$ yields

$$0 \le t_{N+1} - \frac{2}{\pi} \le \frac{1}{\pi N}.$$

For any $\varepsilon > 0$, a positive integer index $n > 1 + 1/(\pi\varepsilon)$ has $N = n - 1 > 1/(\pi\varepsilon)$. Therefore

$$\left| t_n - \frac{2}{\pi} \right| \le \frac{1}{\pi(n-1)} < \varepsilon.$$

This proves the conjectured limit by Definition 7.1.

Conclusion. Part (a) proves existence using Section 10 alone. For part (b), the computed terms suggest $2/\pi$. The explicitly marked additional calculus argument verifies that

$$\boxed{\lim_{n\to\infty} t_n = \frac{2}{\pi}.}$$

No product formula was taken on trust; the finite-product identity and its limiting estimate were derived.

10.12 Exercise 10.12: A product recurrence with an exact formula

What we must show. For

$$t_1 = 1, \qquad t_{n+1} = \left(1 - \frac{1}{(n+1)^2}\right) t_n \quad (n \in \mathbb{N}),$$

(a) establish existence of the limit, (b) guess its value, (c) prove $t_n = (n+1)/(2n)$ by induction, and (d) use the formula to verify the limit.

Part (a), Step 1: prove positivity and monotonicity.

For every $n \geq 1$, we have $n+1 \geq 2$, so

$$0 < \frac{1}{(n+1)^2} \leq \frac{1}{4}, \qquad \frac{3}{4} \leq 1 - \frac{1}{(n+1)^2} < 1.$$

The initial term is positive. If $t_n > 0$, multiplying it by this positive coefficient gives $t_{n+1} > 0$. Induction proves positivity for every term.

Because the coefficient is strictly less than 1 and $t_n > 0$, we also have $t_{n+1} < t_n$ for every n. Thus the sequence is strictly decreasing, bounded below by 0 and above by $t_1 = 1$.

Part (a), Step 2: apply bounded monotone convergence.

Theorem 10.2 now gives a finite limit L. Exercise 8.9 implies $0 \leq L \leq 1$.

Conclusion. The limit exists as a real number. Its value has not been assumed in establishing existence.

Part (b): calculate early terms to formulate a guess.

Successive substitutions give

$$t_2 = \left(1 - \frac{1}{4}\right) 1 = \frac{3}{4},$$

$$t_3 = \left(1 - \frac{1}{9}\right) \frac{3}{4} = \frac{8}{9} \cdot \frac{3}{4} = \frac{2}{3},$$

$$t_4 = \left(1 - \frac{1}{16}\right) \frac{2}{3} = \frac{15}{16} \cdot \frac{2}{3} = \frac{5}{8},$$

and

$$t_5 = \left(1 - \frac{1}{25}\right) \frac{5}{8} = \frac{24}{25} \cdot \frac{5}{8} = \frac{3}{5}.$$

The values

$$1, \quad \frac{3}{4}, \quad \frac{2}{3}, \quad \frac{5}{8}, \quad \frac{3}{5}$$

fit the proposed pattern $t_n = (n+1)/(2n)$. Since this expression equals $1/2 + 1/(2n)$, it suggests the limit $1/2$. At this stage, the pattern and the limit are guesses to be proved.

Part (c), Step 1: state the induction proposition and verify its base.

Let P_n be the assertion

$$t_n = \frac{n+1}{2n}.$$

For $n = 1$, the right side is $(1+1)/(2 \cdot 1) = 1$, which equals t_1. Thus P_1 holds.

Part (c), Step 2: substitute the induction hypothesis.

Assume P_n for an arbitrary positive integer n. The recurrence gives

$$t_{n+1} = \left(1 - \frac{1}{(n+1)^2}\right) \frac{n+1}{2n}$$

$$
\begin{aligned}
&= \frac{(n+1)^2 - 1}{(n+1)^2} \frac{n+1}{2n} \\
&= \frac{n^2 + 2n}{(n+1)^2} \frac{n+1}{2n} \\
&= \frac{n(n+2)}{(n+1)^2} \frac{n+1}{2n} \\
&= \frac{n+2}{2(n+1)} \\
&= \frac{(n+1)+1}{2(n+1)}.
\end{aligned}
$$

The cancellation in the penultimate line is legitimate because $n > 0$ and $n+1 > 0$, so neither cancelled factor is zero. The final expression is precisely the formula in P_{n+1}.

Conclusion. Mathematical induction proves

$$\boxed{t_n = \frac{n+1}{2n} \quad \text{for every } n \in \mathbb{N}.}$$

Part (d), Step 1: separate the constant term from the error.

The proved formula gives the identity

$$t_n = \frac{n+1}{2n} = \frac{n}{2n} + \frac{1}{2n} = \frac{1}{2} + \frac{1}{2n}.$$

Since $1/n \to 0$, the limit laws from Section 9 give $t_n \to 1/2$.

Part (d), Step 2: verify the conclusion directly with a threshold.

For any $\varepsilon > 0$, choose $N = 1/(2\varepsilon)$. Every positive integer $n > N$ satisfies $1/(2n) < \varepsilon$, and

$$\left| t_n - \frac{1}{2} \right| = \left| \frac{1}{2n} \right| = \frac{1}{2n} < \varepsilon.$$

This is the definition of convergence to $1/2$.

Conclusion. The guess in part (b) is now proved:

$$\boxed{\lim_{n \to \infty} t_n = \frac{1}{2}.}$$

The exact formula also gives the error $t_n - 1/2 = 1/(2n)$ for every positive integer index.